MODELLING SURVIVAL THROUGH COMPOSITION OF REINFORCEMENT LEARNING POLICIES

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June 8, 2026

A dissertation submitted to the Faculty of Science, University of the Witwatersrand, Johannesburg, in fulfilment of the requirements for the degree of Master of Science

Abstract

Traditionally, the objective of reinforcement learning (RL) is to determine a policy that maximises cumulative rewards. Recently, however, new approaches have emerged in which agents prioritise avoiding terminating states over reward maximisation. This paradigm, often referred to as survival reinforcement learning, shifts the focus from expected return to sustained operation in hazardous environments.

Several studies have explored survival strategies for RL agents whose primary goal is to remain alive by avoiding dangerous states rather than maximising cumulative reward. For example, some works propose methods that explicitly focus on avoiding terminal states, while others formulate survival as maximising multi-step survival probability through carefully designed reward functions. In both cases, the survival task is typically learned from scratch using standard RL techniques such as policy gradient methods or value-based approaches.

In this paper, we advance the field by formulating survival as a composition of previously learned base tasks rather than treating it as an entirely new challenge. Our results demonstrate that survival agents benefit significantly from this compositional approach because they learn survival behaviours more quickly by leveraging existing skills and achieve superior survival performance compared to agents trained to learn the full survival task from scratch.

This compositional perspective highlights the efficiency and adaptability of survival-oriented RL, showing that agents can generalise survival strategies by reusing and recombining prior knowledge. We demonstrate that a survival agent not only learns to survive more quickly when it composes its existing skills, but also achieves superior survival performance compared to learning the full survival task from scratch.

Declaration

I, *Caston Itayi Nyabadza*, hereby declare the contents of this dissertation to be my own work. This dissertation is submitted for the degree of Master of Science in Computer Science at the University of the Witwatersrand. This work has not been submitted to any other university or for any other degree.

Acknowledgements

I wish to sincerely acknowledge the invaluable support I received throughout my MSc in Computer Science research studies. I am deeply grateful to my wife and children for their patience and encouragement, to my supervisors for their guidance and expertise, and to my colleagues at work for their understanding and assistance. Their collective support has been instrumental in the successful completion of this journey.

08/06/26

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Chapter 1

Introduction

Imagine a robotic puppy that is lost in the woods and needs to survive for as long as possible, but it is exposed to ever-changing environments which are both dangerous and harsh (Figure 1.1). In other words, it needs to improve its life expectancy. For it to survive, it needs to forage for food and water and get some rest along the way. In contrast to the normal Reinforcement Learning (RL) problem that requires maximising the cumulative rewards [Sutton and Barto 2018], Survival-Oriented Reinforcement Learning (SORL) prioritises survival first [Ye *et al.* 2017].

Since such a problem requires an agent to live longer, it poses the following research challenges: (1) a strong need for generalisation, (2) re-usability of skills, and (3) evaluation of long-term memory and reasoning [Hafner 2021]. Such research challenges are usually tackled through Deep Reinforcement Learning (DRL) due to its ability to learn over high-dimensional observations (like images) and still achieve good performance. However, DRL is susceptible to noise and has challenges in exploring the state-action space; hence, it can be easily trapped in a local maximum [Ye *et al.* 2017]. Currently, SORL agents learn to survive by learning everything from scratch until they reach the best life expectancy they can achieve. During learning, the agent must try another strategy if the current policy has reached a maximum number of steps, with the hope that the new policy has more steps than the previous strategy [Ye *et al.* 2017].

Nangue Tasse *et al.* [2020] proved that it is more efficient to compose existing skills to solve a more complex task than to learn the complex task from scratch. Likewise, we intend to use the composition of existing survival sub-skills to determine the optimal policy that provides the SORL agent with the longest life expectancy.

To understand the necessity of composition, consider the delicate balance the puppy must maintain. It is rare that the environment provides a single resource without introducing new constraints; see Figure 1.1. For example, the puppy may be dangerously thirsty but already comfortably full. Navigating to a resource that offers an undifferentiated mix of food and water might quench its thirst but simultaneously push it into a lethal "overfed" state. Consequently, the puppy cannot simply trigger one behaviour in isolation; it must learn to intelligently blend its base skills—eating, drinking, and resting—to address multi-faceted internal needs. At each step, the puppy evaluates its

context and internal state to determine the required skill composition. Managing the survival needs effectively depends entirely on its current physical state. When it is both *thirsty* and *hungry*, the primary objective is to locate resources that provide a dual benefit of *hydration* and *sustenance* (Water *AND* Food). However, if the puppy finds itself *thirsty* but *full*, it must focus exclusively on *water intake*; in this state, it is crucial to *avoid further food* to prevent the risks associated with overfeeding (Water *AND NOT* Food).

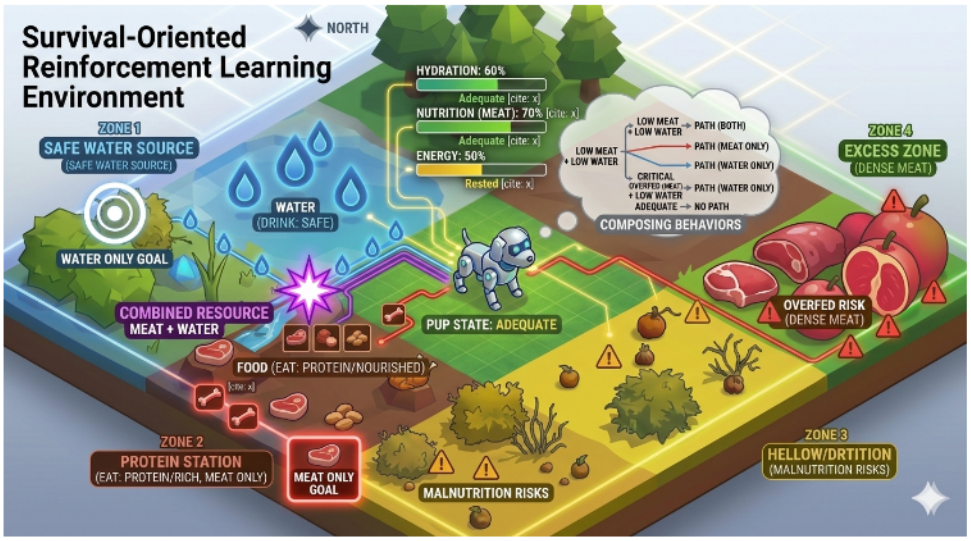


Figure 1.1: *The SORL Survival Environment.* This visualisation illustrates the puppy navigating a complex landscape. To survive, it must evaluate its internal meters (Hydration, Nutrition, Energy) and compose actions to avoid hazards like "Malnutrition Zones" or "Excess Zones" while seeking "Safe Water" and "Protein Stations". AI-generated artwork created with Gemini AI."

In this SORL environment, survival is defined by adequacy, which is a precise window of biological success where the puppy is neither deficient nor overloaded. The puppy must compose the skills it was "born" with or prior learnings to reach this state. The boundaries of adequacy are strict because the puppy must be neither too thirsty nor "drowning" from excess water, and neither hungry, nor overfed. Failure to meet any of these specific needs causes the robotic puppy to die, resulting in the immediate termination of the episode as a failed attempt.

The concept of behaviour composition for survival can be extended to industrial and automotive agents by defining "adequacy" as a set of critical operational thresholds that must be balanced simultaneously. For a warehouse robot, survival could depend on composing energy-efficient movement with thermal management; if it ignores task demand, it fails on utility, but if it ignores its internal temperature, it faces hardware "death" via overheating. Similarly, a self-driving car in a harsh environment must compose traction control, fuel conservation, and sensor-driven hazard avoidance to improve its life expectancy; for instance, if it is low on fuel during a storm, it cannot simply accelerate to a station (risking a crash) but must blend a lower rev-per-minute "Conserve" skill with a high-grip "Stability" skill. In both cases, the agent moves beyond simple

reward maximisation to a SORL approach, where the optimal policy is a continuous composition of sub-skills like power management, mechanical preservation, and environmental navigation tailored to the current context to avoid lethal failure states.

The critical failure of these automated SORL agents presents a high-stakes risk, potentially leading to the loss of human life or catastrophic, costly damage to the hardware itself. While modern agents are typically trained from the ground up using DRL methods, this "from scratch" approach is inherently inefficient [Mnih *et al.* 2015]. It fails to capitalise on the re-usability of existing skills such as basic navigation or resource consumption, which could otherwise be composed to more effectively teach the agent how to extend its life expectancy in harsh, ever-changing environments. We are proposing to compose the base skills to train an agent to survive longer than it would have if it were taught from scratch all the time. Furthermore, as environmental configurations or task demands shift, the proposed framework is designed to allow previously composed target tasks to function as modular components for higher-level combinations, offering a structural pathway to potentially reduce the computational resources and training horizons required to adapt to novel environmental demands.

The objective of this research is to develop and evaluate a Hierarchical-Compositional Survival Reinforcement Learning (HCSRL) framework that enables agents to survive longer in dynamic environments by reusing and composing primitive survival skills. Specifically, this study investigates how skill composition can improve generalisation, reduce training costs, and enhance long-term survival compared to agents trained from scratch or those relying solely on goal-specific options. The contributions of this work are threefold: (1) the design of a survival-oriented RL framework that leverages compositional skill reuse, (2) a comparative analysis of HCSRL against Goal-Oriented RL and Q-learning survival agents across varying goal settings, and (3) empirical evidence demonstrating that under the evaluated asset densities, state dimensions, and resource decay constraints, compositional strategies yield higher early-stage survival rates, extended episode durations, and improved resource allocation profiles. Together, these contributions advance the understanding of survival-oriented reinforcement learning and provide a foundation for scalable applications in safety-critical domains.

This thesis demonstrates that HCSRL, through the reuse and composition of primitive survival skills, enables agents to achieve longer life expectancy and higher early-stage sample efficiency within the parameterised bounds of the evaluated multi-resource environments compared to traditional flat and hierarchical baselines.

The structure of this dissertation is organised as follows. Chapter 2 introduces the fundamental concepts of Reinforcement Learning (RL), outlines the various RL approaches upon which this work is built, and provides an overview of current research in the field. Chapter 4 presents the methodology, detailing the design of our Hierarchical-Compositional Survival Reinforcement Learning (HCSRL) agent and explaining how survival-based tasks are composed to extend life expectancy. Chapter 5 and 6 describe the experimental setup, evaluate the performance of our framework against alternative methods, and analyse the results. Finally, Chapter 7 summarises the key contributions of this work and discusses potential directions for future research.

Chapter 2

Background

This chapter introduces the fundamental concepts of Reinforcement Learning (RL) and the Q-learning algorithm, which form the foundation of our work. We then expand to other RL approaches that underpin our framework. Section 2.2 discusses Goal-Oriented RL, a variant of the standard RL problem that incorporates explicit goal states. In our context, these goal states represent survival-based tasks. Section 2.3 introduces Compositional RL, which addresses complex RL problems by composing primitive tasks. This concept is central to our approach, as it enables agents to reuse and combine survival skills rather than relearning them from scratch. Section 2.4 explores Survival and Safety-Oriented RL, which differs from traditional RL by prioritising survival and/or safety over cumulative reward maximisation. This principle is essential for HCSRL, where the agent's objective is to maximise life expectancy under dynamic and potentially hazardous conditions. Together, these strands of research provide the theoretical and methodological foundation for the Hierarchical-Compositional Survival RL framework developed in this dissertation.

2.1 Reinforcement Learning

RL involves an agent interacting with an environment to learn to choose actions that contribute towards its desired goal. Environments are often formulated as Markov Decision Processes (MDPs), characterised by the tuple $M = \langle S, A, p, r, \gamma \rangle$. Here, S is the state space that represents all possible states of the environment. The action space, A, is all the possible actions the agent can execute in a given state. The transition probability, p, is the probability that the agent will move from one state to another under a given action. r is the immediate reward received after executing action a in state s. The discounting factor $\gamma \in [0, 1]$, discounts rewards exponentially the further they are into the future [Sutton and Barto 2018].

The agent's policy π determines which action $a \in A$ is taken when in any observed state $s \in S$. The optimal policy maximises the accumulated reward by the agent over time. During training, at every step, the agent observes a state s, selects an action a according to the policy π, executes it and receives a reward r. To gauge the quality of

the agent being in the state $s \in S$ or choosing action $a \in A$ when in the state $s \in S$, it is important to assign a value to that state or state-action pair. State or state-action values are represented by state-value functions and state-action value functions (Q-functions), respectively. The agent typically wishes to maximise the expected return G_t, representing the sum of discounted rewards received by an agent starting at time-step t. The state-value function $V_\pi(s_t)$ is simply the expected total return G_t when beginning in state s_t and using policy π to select actions from there onwards (see Equation 2.1).

$$V_\pi(s_t) \;=\; \mathbb{E}[G_t|S = s_t] \;=\; \mathbb{E}_\pi\left[\sum_{t=0}^{\infty} \gamma^t r_t | S = s_t\right] \tag{2.1}$$

Similarly, the state-action value or Q-function is the cumulative reward G_t for choosing action a_t when in state s_t and thereafter following policy π (see Equation 2.2).

$$Q_\pi(s_t, a_t) \;=\; \mathbb{E}_\pi\left[\sum_{t=0}^{\infty} \gamma^t r_t | S = s_t, A = a_t\right] \tag{2.2}$$

These value functions are subsequently used to organise and structure the search for better policies [Sutton and Barto 2018]. By definition, a policy π_2 is better or equal to π_1 if and only if the expected returns are greater than or equal to those of π_1 for all the states. There is always at least one optimal policy π_*, that is better than or equal to all the other policies [Sutton and Barto 2018]. It is, therefore, possible that multiple optimal policies exist but they will all have the same optimal value function because policy π_* will have a value function $V_* \geq V_{i\forall n}$ for all $s \in S$ (see Equations 2.3 and 2.4).

$$V_*(s) \;=\; \max_\pi V_\pi(s) \tag{2.3}$$

for all $s \in S$.

$$Q_*(s, a) \;=\; \max_\pi Q_\pi(s, a) \tag{2.4}$$

for all $a \in A$ and $s \in S$.

2.1.1 Q-learning

Q-learning is a model-free reinforcement learning algorithm, i.e. it does not require prior knowledge of the environment, and is designed to help an agent learn the optimal actions to take in any given state to maximise rewards. Q-learning uses the dynamic programming technique because (1) it can be decomposed into smaller sub-problems and (2) these sub-problems recur multiple times, hence it can re-use the already stored solution to calculate and update a new solution of the Q-value [Sutton and Barto 2018]. This method does value iteration by recursively doing the policy evaluation, i.e. one update for each state s followed by a greedy improvement step for every sweep [Jang *et al.* 2019].

This RL algorithm makes use of a data structure called the Q-table, which stores the Q-values, i.e. expected future rewards for each action-state pair $Q(s, a)$. The Q-table is initialised with arbitrary values, usually zeros. The agent interacts with the environment, taking actions based on the Q-table. It balances taking random actions to gather information (exploration) and choosing actions with the highest Q-values to maximise rewards (exploitation). The agent then observes the next state $s^{'}$ and receives a reward r based on the action a taken.

By removing the expectation sign in the action-value function in equation 2.2, $Q(s, a)$ can be expressed as the following non-linear Bellman equation 2.5. By being an off-policy method, Q-learning separates the acting policy from the learning policy [Jang *et al.* 2019]. It does not use the Q-value of the action a selected by the acting policy to update the policy, but greedily selects the highest Q-value. This becomes the best estimate of the Q-value at any given time, which the agent continuously learns by bootstrapping from Temporal Difference.

$$Q(s, a) = \sum_{s^{'}, r} p(s^{'}, r|s, a) \left[r(s, a) + \gamma \max_{a^{'}} Q(s^{'}, a^{'}) \right] \tag{2.5}$$

Q-learning assumes that certain dynamics of the MDP are known up-front, especially the reward r, hence to reduce computational complexity, they are given up-front to the agent. The transition probability p is usually calculated by an incremental mean of the number of times the agent has been in a state $\frac{1}{N_{s_t}}$, or in non-stationary problems, it might be better to ignore old episodes and replace this with a learning rate α. The agent refines its state–action value function by repeatedly applying the update rule to samples drawn from the environment, refer to Equation 2.6.

$$\underbrace{\text{New } Q(s, a)}_{\substack{\text{New} \\ \text{Q-Value}}} = \underbrace{Q(s, a)}_{\substack{\text{Current} \\ \text{Q-Value}}} + \alpha \Big[\underbrace{r(s, a)}_{\text{Reward}} + \gamma \overbrace{\max Q'(s', a')}^{\substack{\text{Maximum predicted reward, given} \\ \text{new state and all possible actions}}} - Q(s, a) \Big] \tag{2.6}$$

The value iteration will continue until the temporal difference error, i.e. the difference between the current estimate and the last estimate, is very small. The agent would have learned the optimal policy, see algorithm 1.

Algorithm 1 Q-learning

Input: policy π, learning rate α, discount factor γ, exploration rate ϵ, max episodes E
Initialise the Q table arbitrarily for all state-action pairs
for *all* $i \in [1, ...,\text{E}]$ **do**
  Observe initial state $s_0 \in \mathcal{S}$
  while $s \neq$ terminal state **do**
    Calculate π according to Q and ϵ-greedy exploration strategy
    Choose action a using policy π
    Observe reward r and new state s'
    $Q(s,a) \leftarrow (1-\alpha) \cdot Q(s,a) + \alpha \cdot (r + \gamma \cdot \max_{a'} Q(s',a'))$ ▷ Update the $Q(s,a)$
    $s \leftarrow s'$
return Q

The challenge with Q-learning is that it works well with finite and discrete action and state space problems. As the state and action space increase, it would need a larger Q-table, which increases memory requirements, and the convergence speed decreases. Since it also most likely won't visit certain states often, it might be very difficult to generalise the value function.

2.1.2 Taxonomy of Value Updates: Model-Free vs. Model-Based Paradigms

To properly contextualise the tabular algorithms utilised throughout this work, it is mathematically vital to classify classical Q-learning within the broader reinforcement learning taxonomy. Classical Q-learning is a strictly model-free temporal difference (TD) architecture. It estimates optimal action-values directly from experienced sample trajectories without constructing, learning, or querying an explicit representation of environmental transitions or reward physics [Sutton and Barto 2018]. The update operator relies solely on raw sample transitions as per Equation 2.5.

Conversely, model-based formulations explicitly approximate or possess prior access to the true transition dynamics $P(s' \mid s,a) = \mathbb{P}(S_{t+1} = s' \mid S_t = s, A_t = a)$ and reward distributions $R(s,a)$ [Sutton and Barto 2018]. A model-based agent leverages these estimated structures to perform background planning updates—such as through value iteration or tree-search rollouts—by taking expectations over all prospective next-states without executing physical primitives in the actual environment, see Equation 2.7.

$$Q(s,a) = R(s,a) + \gamma \sum_{s' \in S} P(s' \mid s,a) \max_{a'} Q(s',a') \tag{2.7}$$

Although model-based methods offer superior sample efficiency through simulated experiences [Sutton and Barto 2018], they suffer heavily from compounding model errors and steep computational overhead in highly expansive or non-linear transition spaces [Janner *et al.* 2019; Lai *et al.* 2020]. To circumvent these limitations, the framework presented in this dissertation retains a model-free update scheme at the primitive level.

By bypassing the burden of explicit transition modelling, the proposed approach leverages high-level hierarchical abstractions to recover sample efficiency [Nachum *et al.* 2018].

2.2 Goal-oriented Reinforcement Learning

In Goal-Oriented RL (GoRL), the agent is trained to solve a sequence of related tasks and to learn policies that enable it to reach multiple goals [Zhu *et al.* 2021]. Unlike the normal RL setup, GoRL enables the agent to achieve a set of predefined goals. Most GoRL problems are episodic, and the RL agent has to achieve the goal states within an environment while maximising the cumulative reward or minimising the cumulative cost [Tarbouriech *et al.* 2020]. The primary objective of GoRL is to learn an optimal policy that maps states to actions in order to achieve defined goals with minimal cumulative cost in an unknown environment. This process is typically modelled as an episodic Stochastic Shortest Path (SSP) problem, where each episode begins at a fixed initial state s_0, and the agent sequentially selects actions, incurs costs, and transitions between states until reaching the goal state [Chen and Luo 2022].

The GoRL MDP is defined as the tuple $M = \langle S, A, p, G, r_g, \gamma, \phi, \rho_0, p_g \rangle$, where G denotes the goal space, $\phi : S \to G$ is a tractable mapping specifying goal representation, ρ_0 is the initial state distribution, and p_g is the distribution over desired goals. The reward function r_g is sparse and goal-conditioned, providing a binary signal upon goal achievement [Zhu *et al.* 2021; Ren *et al.* 2019]; see Equation 2.8.

$$r_g(s_t, a_t, s_{t+1}) = \begin{cases} 0, & ||\phi(s_{t+1}) - g||_2 \leq \delta_g \\ -1, & \text{otherwise} \end{cases} \tag{2.8}$$

δ_g is a given threshold indicating whether the goal is considered to be reached [Ren *et al.* 2019].

Similarly, GoRL is defined as a universal value function that employs function approximation to represent a family of goal-conditioned value functions, refer to Equation 2.10.

$$V_\pi(s_t, g) = \mathbb{E}_{\substack{a_t \sim \pi \\ s_{t+1} \sim p}} \left[\sum_{t=0}^{\infty} \gamma^t r_g(s_t, a_t, s_{t+1}) | S = s_t \right] \tag{2.9}$$

GoRL aims to derive a goal-conditioned policy $\pi(a|s, g)$ that maximises the expected discounted cumulative reward over the distribution p_g [Zhu *et al.* 2021]:

$$\eta = \mathbb{E}_{\substack{a_t \sim \pi \\ s_{t+1} \sim p \\ g \sim p_g}} \left[\sum_{t=0}^{\infty} \gamma^t r_g(s_t, a_t, s_{t+1}) | S = s_t \right] \tag{2.10}$$

2.3 Compositional Reinforcement Learning

Composition allows the already learned skills to be combined to solve a new task without explicitly learning to solve it. It allows multiple primitive policies used to solve specific tasks to be combined to create an advanced policy that concomitantly solves a more complex task [Haarnoja *et al.* 2018]. The main aim of compositionality is to use the knowledge obtained in previous tasks to solve any new skill without the agent explicitly learning that new task, which is a property known as zero-shot task generalisation [van Niekerk *et al.* 2018; Nangue Tasse *et al.* 2020]. Composition takes basic policies, passes them to more complex problems, and adeptly learns a composite policy through RL [Qureshi *et al.* 2019].

Being able to compose makes intelligent agents capable of learning new abilities from a combination of existing skills, and this provides them with a combinatorial explosion of all the possible tasks they can perform [Nangue Tasse *et al.* 2020]. Once an agent has explicitly learned a set of base skills, it can make them solve an exponential number of new tasks without the need to explicitly learn these newer tasks [Nangue Tasse *et al.* 2020]. This gives the agent the ability to perform lifelong learning as it will act efficiently when presented with new and unseen tasks [van Niekerk *et al.* 2018]. Therefore, it is more efficient for an agent to learn new behaviours by composing existing skills compared to training from scratch [Haarnoja *et al.* 2018].

Haarnoja *et al.* [2018]; van Niekerk *et al.* [2018] discuss logical composition in RL primarily through the union and intersection of tasks, but they stop short of giving formal definitions. Their work tends to treat these operations more as intuitive constructs, i.e. ways of combining reward functions or policies instead of rigorously defined mathematical objects.

Nangue Tasse *et al.* [2020] formalised the tasks as a Boolean Algebra by defining a task through its reward function $r(s, a, s')$. By viewing tasks as sets of desirable outcomes, they formalised the composition of tasks using the three fundamental operators of Boolean algebra: (1) Disjunction (OR/ $\vee$) which seeks to achieve at least one of the specified tasks, (2) Conjunction (AND / $\wedge$) which simultaneously satisfies the requirements of multiple tasks, and (3) Negation (NOT / $\neg$) which avoids the goals of a specific task while still operating within the environment.

The primary technical hurdle in composing value functions was the fact that standard Q-functions ($Q(s, a)$) only track the value of the immediate next goal. To overcome this, Nangue Tasse *et al.* [2020] introduce Extended Value Functions. These functions represent a form of "mastery" over the environment by encoding the value of reaching any state from any other state. This global information is essential because a state that is suboptimal for Task A might be the optimal starting point for the conjunction of Task A and Task B. The extended Q-value, which is used to learn the base tasks, is, however, a modification of a normal Q-value. The difference is that the extended Q-value keeps track of all the terminating states, i.e., the goals $\forall g \in G$, and at each time-step t, it updates with regard to the current state s_t, action a_t taken, and all the goals already encountered [Nangue Tasse *et al.* 2020]. Nangue Tasse *et al.* [2020]

defined the extended reward function $\bar{r}$, refer to Equation 2.11, and extended Q-value function $\bar{Q}$, refer to Equation 2.12.

$$\bar{r}(s,g,a) \;=\; \begin{cases} \bar{r}_{MIN}, & g \neq s \in G \\ r(s,a), & \text{otherwise} \end{cases} \tag{2.11}$$

Where $\bar{r}_{MIN} \leq \min\{r_{MIN}, (r_{MIN} - r_{MAX})D\}$, and D is the diameter of the MDP [Nangue Tasse *et al.* 2020].

$$\bar{Q}^{\bar{\pi}}(s,g,a) \;=\; \bar{r}(s,g,a) \;+\; \int_S \bar{V}^{\pi}(s',g)\rho(s,a)(ds') \tag{2.12}$$

Where $\bar{V}^{\bar{\pi}}(s,g) \;=\; \mathbb{E}_{\pi}\left[\sum_{t=0}^{\infty} \bar{r}(s_t,g,a_t)\right]$ [Nangue Tasse *et al.* 2020].

The core theoretical contribution is the proof of a homomorphism between the algebra of tasks and the algebra of value functions. Nangue Tasse *et al.* [2020] demonstrate that under deterministic transition dynamics, the optimal value function for a composed task can be directly obtained from the value functions of its components via point-wise operations, see Equations 2.13, 2.14 and 2.15, which is a specific transformation of the mastered value function.

$$Q_{A \vee B}{}^{*}(s,g,a) = \max(Q_A{}^{*}(s,g,a), Q_B{}^{*}(s,g,a)) \tag{2.13}$$

$$Q_{A \wedge B}{}^{*}(s,g,a) \;= \min(Q_A{}^{*}(s,g,a), Q_B{}^{*}(s,g,a)) \tag{2.14}$$

$$Q^{*}_{\neg A}(s,g,a) \;= (Q_A{}^{*}(s,g,a) + Q_B{}^{*}(s,g,a)) - Q_A{}^{*}(s,g,a) \tag{2.15}$$

The number of primitive survival skills required to solve arbitrary tasks in a multi-goal environment increases only logarithmically with the number of goal states [Nangue Tasse *et al.* 2020]. Formally, an environment with n distinct goals requires the agent to learn approximately $k = \log_2(n) + 1$ base tasks. By learning a small set of k base tasks, an agent can immediately solve any of the 2^{2^k} tasks possible within that Boolean algebra. This exponential growth demonstrates that a relatively small collection of primitive skills can support an astronomically large repertoire of zero-shot task combinations, substantially enhancing the agent's adaptability and generalisation in previously unseen environments. This provides a path toward agents that can understand and execute complex, high-level instructions—such as "Collect all red items AND all circles, but NOT if they are small"—instantly, provided they have mastered the base concepts of "red," "circles," and "small."

2.4 Survival Reinforcement Learning

In Survival RL (SRL), the agent's main goal is to survive, unlike normal RL, where the main goal is to accumulate maximum rewards [Moazami and Doerschuk 2020].

In this type of RL, survival is in itself a learning objective, not a means to an end to another defined goal. Instead of maximising cumulative rewards, the agent focuses on avoiding terminal or dangerous states. This type of RL replaces traditional reward maximisation with a survival-oriented strategy. Unlike standard RL, which typically seeks to maximise an arbitrary cumulative reward, SRL focuses on the objective of persistence; the primary goal of the agent is to avoid a terminal "failure" or "death" state for as long as possible. In classical RL, an agent's objective is defined by the Bellman equation to maximise the expected sum of discounted rewards $G_t = \sum_{k=0}^{\infty} \gamma^k r_{t+k+1}$. In contrast, SRL often redefines the objective as the maximisation of the agent's lifespan and/or the minimisation of a hazard function.

In survival analysis, the agent's "risk" at any time step can be modelled via a hazard rate $\lambda(t)$, which represents the probability of failure given survival up to time t. The RL agent's policy π is optimised to maximise the survival function $S(t) = P(T > t)$, where T is the random variable representing the time of failure. The most prominent application of SRL is in Dynamic Treatment Regimes (DTRs) within healthcare. In this context, the "agent" is a clinical decision-support system, the "environment" is the patient's physiological state, and the objective is to maximise the lifespan and/or safe exploration. For example, in oncology or chronic disease management, the reward is often a surrogate for clinical outcomes such as disease remission or overall survival.

Murphy [2003] pioneered using RL to optimise the sequence of treatments (e.g., chemotherapy doses or psychiatric interventions) to maximise the time until a negative health event occurs. This work highlights a critical challenge in SRL, especially for medicine, is the "safety-survival" constraint. Unlike game-playing AI, a clinical agent cannot afford "trial-and-error" that leads to patient death. This has led to the development of safety-survival RL algorithms that incorporate constraints into the Markov Decision Process (CMDP) to ensure the agent stays within a "safe envelope" of states [Wagener *et al.* 2021; Banumathi 2025; Basu *et al.* 2025]. The main aim of safety-survival RL (Safe RL) is to learn optimal policies while staying safe [Yang *et al.* 2023]. This ensures that the learning agent learns an optimal policy while ensuring safety constraints [Wagener *et al.* 2021]. Most Safe-survival RL studies decouple the safety aspect from the reward aspect using CMDPs, where an independent signal models the safety aspect and the agent needs to autonomously find a trade-off between performance and safety [Simão *et al.* 2021; Wagener *et al.* 2021; Liu *et al.* 2020]. In this CMDP, the agent interacts with an environment without full knowledge about the transition, reward, and cost functions [Simão *et al.* 2021].

In this CMDP, the agent interacts with an environment without full knowledge about the transition, reward, and cost functions [Simão *et al.* 2021]. We still consider an MDP with a state space S, an action space A, a transition function ρ that represents the conditional probability distribution $\rho(s'|s, a)$ of moving to a successor state $s' \in S$ after executing action $a \in A$ in state $s \in S$, a bounded reward function $r : S \times A \to [0, 1]$, and an initial state distribution d_0.

The CMDP, however, considers a safety indicator or cost function $c(s'|s, a)$ which is an indicator of being in an unsafe state [Wagener *et al.* 2021]. This CMDP is characterised by the tuple $\langle S, A, \rho, c, r, \gamma \rangle$ which uses a reward-based MDP $M = \langle S, A, \rho, r, \gamma \rangle$ and a

cost-based MDP $\bar{M} = \langle S, A, \rho, c, \gamma \rangle$ [Wagener *et al.* 2021]. The survival-safe RL agent must find a policy with the maximum expected reward among the safe policies, namely those with expected cost smaller than a safety threshold. The survival-safe RL can therefore be defined as a constraint policy optimisation problem that aims to find a policy that has a high cumulative reward $V^{\pi}(d_0)$ with cumulative cost $\bar{V}^{\pi}(d_0)$ below the allowed failure probability δ from any initial state distribution d_0 (see Equation 2.16) [Wagener *et al.* 2021]. Therefore, the agent must find a policy with the maximum expected reward among the safe (feasible) policies, namely those with expected costs smaller than a safety threshold.

$$\max_{\pi} V^{\pi}(d_0) \text{ subject to } \bar{V}^{\pi}(d_0) < \delta \tag{2.16}$$

2.5 Hierarchical Reinforcement Learning

For problems characterised by sparse rewards, standard RL agents often struggle with inefficient exploration, potentially wandering the state space without receiving meaningful feedback. Hierarchical Reinforcement Learning (HRL) addresses this by decomposing a complex task into a hierarchy of sub-tasks. In this structure, higher-level policies delegate control to lower-level policies to execute specific sub-goals [Sutton 1998; Sutton *et al.* 1999; Zhang *et al.* 2021]. This introduction of temporal abstraction allows agents to reason over extended sequences of actions, effectively shortening the decision horizon for the high-level controller [Klissarov *et al.* 2025].

A fundamental component of HRL is the Options Framework, which extends the standard MDP action space to include "options"—temporally extended actions that persist over multiple time steps [Sutton *et al.* 1999]. An option ω is typically defined by a triple $(I_\omega, \pi_\omega, \beta_\omega)$, where I_ω is the set of states where the option can be started, π_ω is the intra-option policy that governs the agent's behaviour while the option is active, and the termination condition β_ω is the probability that the option finishes in a given state.

By operating over options rather than primitive actions, HRL reduces the number of decision points required to reach a goal. This enables higher-level policies to benefit from multi-step Temporal Difference (TD) updates, accelerating value propagation over longer temporal horizons [Zhang *et al.* 2021].

Modern HRL often employs a Manager-Worker (or Master-Slave) architecture. The Manager operates at a lower temporal resolution, selecting high-level goals or skills to pursue. The Worker operates at a higher frequency, executing the primitive actions required to satisfy the Manager's current objective. This decomposition facilitates: (1) Structured Exploration by identifying intermediate goals (bottleneck states) that help navigate complex environments, (2) Skill Reusability as learned sub-policies can be invoked across different high-level tasks, promoting transfer learning, and (3) Credit Assignment: Agents can more accurately assign credit across time, as the high-level policy only needs to evaluate the success of an entire option rather than every individual step [Klissarov *et al.* 2025]. These mechanisms collectively allow HRL to solve long-horizon tasks that remain intractable for flat RL architectures.

Chapter 3

Related Work

The development of autonomous agents capable of navigating complex, high-stakes environments requires a paradigm shift from simple reward maximisation to robust, persistence-oriented strategies. This chapter reviews the current landscape of RL as it pertains to three critical pillars of this thesis: goal attainment, logical composition, and survival.

We begin by examining Goal-Oriented RL and its methods for handling sparse feedback. We then explore Compositional RL, which provides the algebraic foundation for combining skills. Finally, we investigate Safe and Survival RL alongside Hierarchical frameworks, identifying the specific technical gaps that necessitate the development of a compositional approach to survival modelling.

3.1 Goal-Oriented Reinforcement Learning

The primary objective of goal-oriented reinforcement learning (GoRL) is to train agents to navigate complex environments to reach specific, multi-target destination states. Unlike traditional reinforcement learning paradigms that optimise a fixed, monolithic reward function, GoRL frameworks typically utilise goal-conditioned policy architectures or Universal Value Function Approximators (UVFAs). Under this formulation, the reward function $r(s, a, s', g)$ and the action-value function $Q(s, a, g)$ are explicitly parameterised by a target goal $g \in \mathcal{G}$, refer to Equation 3.1.

$$Q^*(s, a, g) = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t, s_{t+1}, g) \,\middle|\, s_0 = s, a_0 = a\right] \tag{3.1}$$

While GoRL provides a mathematically flexible foundation for multi-task generalisation, scaling these systems introduces severe optimisation challenges. Because target goals represent a sparse subset of the total state space, agents operating under standard exploration paradigms frequently receive nothing but binary failure signals ($r = 0$), severely hindering value propagation. Current literature addresses these reward sparsity and sequencing bottlenecks through counterfactual trajectory relabelling, predictive model-assisted foresight, and graph-structured hierarchies.

3.1.1 Counterfactual relabelling and Hindsight Experience Replay

To mitigate the sample inefficiency caused by sparse, binary reward signals, a dominant paradigm introduces counterfactual experience manipulation, pioneered heavily by Hindsight Experience Replay (HER) [Andrychowicz *et al.* 2017]. Under the HER framework, when an agent fails to achieve its intended target goal g during an episode, the transition sequence stored in the replay buffer is counterfactually relabelled. The algorithm selects an alternate goal g' that was actually visited during that specific trajectory and substitutes it as the target. By retroactively transforming a behavioural "failure" into a "success" for a different goal configuration, HER forces a dense learning signal out of entirely sparse environments [Andrychowicz *et al.* 2017]. Although highly effective when paired with off-policy algorithms like DQN or Deep Deterministic Policy Gradients (DDPG), HER is fundamentally bottlenecked by its model-free nature. Because it relies strictly on past trajectories, its sample efficiency remains low in expansive environments. Furthermore, it completely lacks goal diversity beyond states that the agent has already physically breached, preventing zero-shot generalisation to unvisited or structurally novel task configurations [Zhu *et al.* 2021].

3.1.2 Model-Assisted Foresight and Dynamics Optimisation

To overcome the experiential constraints of model-free relabelling, subsequent works integrate forward dynamics models to proactively synthesise imaginative goals and supplement real environment interactions, prominent through frameworks such as Model-Assisted Policy Optimisation (MapGo) [Zhu *et al.* 2021]. The MapGo architecture structurally decouples goal generation from past experience via two interlocking mechanisms: Foresight Goal Inference (FGI) and Universal Model-based Policy Optimisation (UMPO). FGI leverages a learned transition dynamics model to project forward and predict future reachable states, using these model-generated states to dynamically relabel trajectories. Simultaneously, UMPO utilises short, branched imaginary roll-outs originating from real environmental states to amplify the density of the training dataset [Zhu *et al.* 2021]. By replacing static historical logs with policy-adaptive, model-generated goals, these methods significantly accelerate sample efficiency. However, their reliability is strictly bound to the accuracy of the forward dynamics model. In highly stochastic or continuous state-action spaces, learned models inevitably suffer from compounding approximation errors (exploding model drift). This drift propagates deceptive value signals through the short-branched roll-outs, introducing severe optimisation instabilities that degrade final policy convergence.

3.1.3 Graph-Structured Hierarchies and Combinatorial Sequencing

When environments require the completion of multiple, highly interdependent long-horizon objectives, flat goal-conditioned architectures become intractable. Recent advancements address this by embedding structural topologies, such as the Graph-Enhanced Hierarchical Reinforcement Learning (GEHRL) framework [Li *et al.* 2023]. These architectures organise multi-goal execution into a distinct two-level decision hierarchy paired with an external relational structure. At the top tier, a high-level agent (e.g.,

optimised via Proximal Policy Optimisation) operates over a coarse temporal scale to plan sequential strings of macro-goals. At the bottom tier, a low-level Actor-Critic agent ingests these macro-goals and executes primitive actions to reach them. To bound the low-level agent's search space, a Graph-based Candidate Selector (GCS) projects the current state onto an external knowledge graph, dynamically filtering out goal-irrelevant states and focusing exploratory behaviour strictly on viable task paths [Li *et al.* 2023]. While graph-structured hierarchies effectively constrain exploration and handle sequential task dependencies, they introduce substantial computational and structural overhead. These systems are heavily dependent on the prior existence or precise construction of a discrete, symbolic knowledge graph. Consequently, they struggle to scale or adapt when deployed in continuous, open-ended environments where target configurations shift fluidly and cannot be neatly captured by rigid topological relations.

3.1.4 Goal-Oriented RL Literature Gaps

Despite the advancements in counterfactual hindsight relabelling, predictive dynamics forecasting, and graph-structured hierarchies, a critical gap remains at the intersection of long-horizon multi-goal navigation and autonomous, physiological survival. The existing GoRL research landscape exhibits three major deficiencies:

- *Static Target Assumption vs. Dynamic Homeostatic Urgency:* Traditional goal-conditioned reinforcement learning frameworks operate under the explicit assumption that target goals are dictated by static, external environmental coordinates or discrete item selections [Andrychowicz *et al.* 2017; Li *et al.* 2023]. They lack the structural mechanisms required to model internal, time-decaying physiological states (S_P). Because standard UVFAs cannot ingest an agent's real-time metabolic deficit trends, they fail to dynamically re-prioritise or re-sequence goals based on changing survival urgencies, executing distant spatial tasks even when the agent is on the verge of catastrophic termination.

- *Absence of Zero-Shot Compositionality in Hindsight Architectures:* While model-assisted methods like MapGo [Zhu *et al.* 2021] expand goal diversity through predictive roll-outs, they remain bounded by the horizons of their forward dynamics models. Current GoRL literature lacks any integration with algebraic skill composition or Boolean task algebras [Nangue Tasse *et al.* 2020]. Consequently, goal-conditioned agents cannot instantly combine learned sub-task primitives zero-shot to satisfy complex combinatorial goal compound configurations (e.g., reaching $Goal_A \wedge \neg Goal_B$), instead requiring entirely new trajectories or expensive model retraining phases for every novel multi-goal variation.

- *Topological vs. Metabolic Constraints:* Graph-based sequencing frameworks, such as GEHRL [Li *et al.* 2023], restrict agent exploration purely by mapping structural relations between external target nodes. These architectures are completely isolated from hard physical boundaries or irreversible absorbing states (ALIVE $=$ FALSE). There remains a distinct lack of frameworks capable of leveraging a high-level sequencing policy to monitor multi-resource metabolic deficits, while

simultaneously navigating an exponentially large, algebraically composed option space ($\Omega = 2^{2n}$) to safely balance survival and external goal achievements.

To resolve these interconnected bottlenecks, this dissertation introduces a novel hierarchical compositional framework. Rather than relying on rigid predefined knowledge graphs or raw, model-free hindsight relabelling, the proposed architecture splits the state space into internal survival levels (S_P) and external positional coordinates (S_E). By utilizing a high-level survival policy (π_s) to monitor physiological deficit trends, the agent can fluidly plan and select optimal survival strategies from a vast, combinatorially rich option space ($\Omega = 2^{2n}$). These strategies are executed zero-shot by a low-level algebraic compositional engine, successfully achieving long-horizon multi-goal navigation while maintaining absolute homeostatic stability.

3.2 Compositional Reinforcement Learning

The primary objective of compositional reinforcement learning is to bypass the sample-inefficiency of training monolithic agents by combining previously learned skills, policies, or value functions to solve novel, complex tasks. Current literature generally bifurcates into algebraic, logical, and structural methodologies.

3.2.1 Algebraic and Value-Based Composition

A dominant paradigm leverages the algebraic properties of value functions to compose policies, pioneered heavily through the integration of Successor Features (SFs) and Generalised Policy Improvement (GPI) [Barreto *et al.* 2018]. Under this framework, environment dynamics are decoupled from reward functions. By maintaining a library of policies Π and evaluating them against a novel reward vector w, agents can instantaneously compute a lower-bound performance guarantee using the generalised policy operator, refer to Equation 3.2.

$$\max_{\pi \in \Pi} Q^{\pi}(s, a) \tag{3.2}$$

This concept has been extended into continuous action spaces using entropy-regularised frameworks (e.g., Soft Q-learning), where individual task-specific Q-functions are linearly combined to approximate optimal behaviour in joint-reward environments [Haarnoja *et al.* 2018; van Niekerk *et al.* 2018]. See Equation 3.3.

$$Q^*_{\Sigma} \approx \sum w_i Q^*_i \tag{3.3}$$

While sample-efficient, algebraic composition fundamentally relies on the assumption of reward additivity or linear transition structures. When tasks exhibit conflicting constraints or non-linear dependencies, linear value-function composition suffers from severe over-estimation bias and destructive interference, failing to guarantee policy safety or optimality.

3.2.2 Logical and Boolean Task Composition

To address the limitations of linear reward additivity, recent works explore formal methods and Boolean logic to compose task specifications. These architectures interpret individual reinforcement learning policies as zero-shot behavioural primitives that can be synthesised using logical operators ($\wedge, \vee, \neg$). Algorithms in this domain map optimal value functions to bounded Boolean spaces, allowing an agent to solve compound tasks like "Achieve Goal A AND NOT Goal B" without further environment interaction [Nangue Tasse *et al.* 2020]. This intersects heavily with Linear Temporal Logic (LTL) and Reward Machines, where automaton states guide the sequential composition of sub-policies [Icarte *et al.* 2018]. The scalability of logical composition is severely bottlenecked by the curse of dimensionality in continuous state-action spaces. Most logic-based CRL frameworks operate within discrete grid-worlds or low-dimensional state representations. When applied to deep, continuous sensory inputs, regularising the value bounds to strictly satisfy Boolean algebraic properties introduces optimisation instabilities that degrade overall performance.

3.2.3 Modular and Parameter-Based Architecture

Rather than composing value functions mathematically, an alternative branch focuses on structural, architectural compositionality within the neural networks themselves. This approach employs modular neural networks, routing mechanisms, or hyper-networks where distinct sub-networks (modules) capture specific facets of an environment's dynamics or policy distributions [Devin *et al.* 2017; Kirsch *et al.* 2018]. Solving a new task becomes an optimisation problem of selecting and routing activations through the correct permutation of pre-trained modules. Modular networks suffer from cross-talk and representation drift. If modules are fine-tuned during the composition phase, they frequently overwrite previously acquired features (catastrophic forgetting). Conversely, if the modules are frozen, the composition layer lacks the expressivity required to adapt to subtle, unmodeled variations in new target domains.

3.2.4 Compositional RL Literature Gaps

Despite the advancements in algebraic, logical, and structural compositional reinforcement learning, a significant gap persists at the intersection of zero-shot task composition and autonomous behavioural survival. Existing CRL frameworks operate under two restrictive assumptions: first, that task objectives are dictated entirely by static, external environmental rewards; and second, that the agent's state representation is monolithic.

This introduces three critical vulnerabilities when applying state-of-the-art Compositional RL to open-ended survival environments:

1. *Neglect of Internal State Dynamics (S_P):* Traditional value-based [Barreto *et al.* 2018] and logical [Nangue Tasse *et al.* 2020] composition methods lack the mechanisms to incorporate homeostatic variables—such as decaying resource levels—that impose hard temporal constraints on the agent's viability. Because flat value com-

positions ($Q^*_\Sigma \approx \sum w_i Q^*_i$) cannot naturally ingest a non-linear, step-by-step internal decay function, they fail to prioritise vital resource replenishment dynamically before catastrophic failure occurs (ALIVE = FALSE).

2. *The Combinatorial Explosion of Bounded Logic:* While Boolean task algebras allow an agent to compose primitives into compound goals zero-shot, the target option space scales exponentially ($k = 2^{2n}$) relative to the number of base environmental tasks n. In flat architectures, evaluating or routing actions through this combinatorial space introduces severe optimisation bottlenecks and policy degradation in continuous or dense environments.

3. *Absence of Hierarchical Orchestration in Zero-Shot Transfer:* Current modular and algebraic paradigms treat composition as a low-level control problem. However, surviving in an ecosystem with shifting internal deficits requires a multi-tier architectural approach. There is a distinct lack of frameworks that decouple the problem into a high-level meta-policy tasked with monitoring physiological survival states (S_P) and selecting broad compositional strategies, while delegating spatial navigation (S_E) to a low-level zero-shot compositional engine.

To address these limitations, this dissertation introduces a novel hierarchical compositional framework. By explicitly bifurcating the state space into internal survival resource levels (S_P) and external positional data (S_E), the proposed architecture uses a high-level survival policy (π_s) to orchestrate a vast, combinatorially rich option space ($\Omega = 2^{2n}$). This decoupling allows the agent to maintain homeostatic stability via meta-level value iteration while exploiting the sample efficiency of zero-shot low-level task composition.

3.3 Safe and Survival-Oriented Reinforcement Learning

The primary objective of safe and survival-oriented reinforcement learning (SORL) is to modify an agent's behaviour to guarantee operational safety, avoid irreversible failure states, or maximise continuous operational lifetime within hazardous environments. Traditional reinforcement learning optimises solely for empirical return, frequently treating catastrophic failures as mere negative reward penalties. In safety-critical or open-ended survival domains, however, architectures must enforce strict transition bounds or prioritise homeostatic longevity. Current literature in this domain generally bifurcates into reactive safety frameworks, monolithic survival optimisation, and multi-resource homeostatic models.

3.3.1 Reactive Safety Frameworks and Constraint Enforcement

A dominant paradigm enforces safety by overlaying runtime monitors, formal verification shields, or predictive state filters on top of an exploratory policy. Prominent methods include Shielding frameworks [Alshiekh *et al.* 2018] and Safe Model-Based Policy Optimisation [Thomas *et al.* 2021]. Under this framework, safety is often formalised via Constrained Markov Decision Processes (CMDPs), where an agent maximises expected return subject to an upper bound δ on expected cumulative cost, see Equation

3.4.

$$\mathbb{E}\left[\sum_{t=0}^{\infty}\gamma^t c(s_t,a_t)\right] \leq \delta \tag{3.4}$$

In runtime shielding architectures, a predictive supervisor (the shield) intercepts the agent's chosen action and dynamically overrides it if it threatens to violate a mathematically verified safety boundary [Alshiekh *et al.* 2018]. Similarly, safe model-based policy optimisation techniques utilise a learned transition model to simulate trajectories ahead of execution, algorithmically pruning actions that exhibit a high probability of stepping into localised hazard states [Thomas *et al.* 2021]. While sample-efficient and robust in preventing immediate physical damage, these frameworks are inherently reactive; they focus heavily on immediate hazard evasion rather than evaluating long-term internal metabolic trends.

3.3.2 Monolithic Survival and Longevity Optimisation

Rather than imposing external boundaries on top of a standard reward structure, survival-oriented reinforcement learning shifts the fundamental objective function from external reward maximisation to the direct maximisation of continuous life expectancy and operational duration [Moazami and Doerschuk 2020]. Under this paradigm, the environment is typically recast as an unrewarded Markov process punctuated by an absorbing terminal state representing agent death or total system failure. The value function is framed to capture the expected time-to-failure or to minimise a cumulative hazard function $H(t)$, refer to Equation 3.5.

$$V^{\pi}(s) = \mathbb{E}_{\pi}\left[\int_0^{\infty} e^{-H(t)}dt \;\middle|\; s_0 = s\right] \tag{3.5}$$

By optimising this objective, the agent learns long-horizon behavioural strategies entirely dedicated to keeping vital system parameters within a survivable envelope [Moazami and Doerschuk 2020]. While these methods excel at sustaining an agent in high-risk environments, they are limited by their monolithic architecture. The learned survival behaviour is tightly coupled to a single, specialised policy trained from scratch, completely lacking the modularity required to decouple, re-use, or compose individual spatial navigation or interaction primitives across novel tasks.

3.3.3 Multi-Resource Homeostasis and Simulation Ecosystems

When agents are deployed in complex real-world settings or long-duration ecosystem simulations, survival cannot be dictated by a single variable; it requires the simultaneous regulation of multiple competing internal and environmental needs [Raza *et al.* 2021; Li and Liu 2024]. These architectures represent the agent's internal state as a multidimensional physiological vector tracking variables such as energy reserves, thermal boundaries, and structural health. The agent's survival relies on maintaining these coupled variables within bounded homeostatic zones. If any single resource falls below its critical threshold, the episode immediately terminates. While these frameworks

capture the intricate dynamics of biological or physical systems, balancing non-linear resource decay channels alongside unpredictable external hazard rates introduces severe scalability and optimisation bottlenecks. Flat networks exposed to these environments frequently suffer from policy instability or exhibit erratic switching behaviour, failing to safely allocate attention across competing urgent needs under strict resource constraints.

3.3.4 Safe and Survival RL Literature Gaps

Despite the advancements in reactive shielding, monolithic longevity optimisation, and multi-resource ecosystem simulations, a significant gap persists at the intersection of safe exploratory execution and zero-shot modular transfer. The existing research landscape reveals several critical deficiencies:

- *Reactive vs. Proactive Persistence:* Most state-of-the-art safety methods, such as Shielding [Alshiekh *et al.* 2018] and Safe Model-Based Policy Optimisation [Thomas *et al.* 2021], are primarily reactive—focused tightly on avoiding immediate, localised unsafe states or recovering from immediate physical damage. There remains a profound lack of research into proactive survival frameworks where the agent's core optimisation driver is the long-term, strategic maximisation of continuous life expectancy through the early forecasting of internal systemic trends.

- *Isolation of Safety and Composition:* While Boolean Task Algebra [Nangue Tasse *et al.* 2020] provides a robust mathematical framework for zero-shot skill composition, it operates in complete isolation from survival metrics like non-linear hazard rates or physiological viability boundaries. Conversely, survival-oriented longevity studies [Moazami and Doerschuk 2020] optimise exclusively for operational lifespan but lack the behavioural modularity and algebraic abstractions required for skill reuse and zero-shot transfer across evolving environment tasks.

- *Scalability in Multi-Resource Environments:* Current survivable robotic frameworks [Raza *et al.* 2021] and artificial ecosystem simulations [Li and Liu 2024] struggle heavily with scalability and computational efficiency when exposed to dynamic environments. When an agent must simultaneously balance multiple independent survival needs (e.g., energy replenishment, health restoration, and dynamic hazard avoidance), flat optimisation architectures suffer from optimisation bottlenecks, often converging to sub-optimal local minima where the agent fails to prevent catastrophic threshold breaches.

To address these limitations, this dissertation introduces a novel hierarchical architecture that bridges the gap between zero-shot composition and multi-resource proactive survival. By integrating a multi-resource homeostatic tracking model (S_P) directly into the high-level policy (π_s), the framework proactively balances competing internal deficit trends. It evaluates safety metrics at every primitive execution step while dynamically selecting optimal behaviours from an algebraically composed, modular option space ($\Omega = 2^{2n}$), successfully achieving long-term survivability without sacrificing skill reuse or optimisation stability.

3.4 Hierarchical and Structured Generalisation

The primary objective of hierarchical reinforcement learning (HRL) is to decompose complex, long-horizon decision-making tasks into a structured arrangement of simpler sub-tasks, thereby improving credit assignment and acceleration of policy generalisation. Formally grounded in the Options Framework and semi-Markov Decision Process (SMDP) formulations, HRL extends the primitive action selection loop to temporally abstract macro-actions, or options $\omega \in \Omega$. Under this formulation, a high-level policy π_s selects an option ω, which executes a low-level policy π_ω over a variable time horizon τ until a termination condition β_ω is satisfied, updating the high-level action-value function as follows:

$$Q^{\pi_s}(s,\omega) = \mathbb{E}\left[\sum_{t=0}^{\tau-1}\gamma^t r_t + \gamma^\tau \max_{\omega'} Q^{\pi_s}(s_\tau,\omega') \,\middle|\, s_0 = s, \omega_0 = \omega\right] \tag{3.6}$$

.

While HRL significantly reduces the effective planning horizon for long-duration problems, discovering, organising, and enforcing safety within these hierarchies remains a formidable Optimisation challenge. Without structured logical abstractions or explicit boundary constraints, multi-tier systems struggle to generalise across shifting environmental topologies or protect the agent from localised physical failure. Current literature attempts to resolve these structuring bottlenecks through logical sub-task graph networks and top-tier state-bounded safe regions.

3.4.1 Logical Dependency Sequencing and Subtask Graphs

To enable zero-shot transfer across complex, interdependent task sequences without manual reward engineering for every permutation, recent paradigms explore structural graph encodings, prominent through architectures such as the Neural Subtask Graph Solver (NSGS) [Sohn *et al.* 2018]. The NSGS framework models multi-step execution by mapping environmental objectives onto a directed acyclic graph representing logical sub-task dependencies. By utilising recursive neural networks (Tree-RNNs), the architecture directly encodes complex precondition strings consisting of Boolean combinations ($\wedge, \vee$). As the agent interacts with the environment, value and reward signals are recursively propagated across the nodes of the dependency graph. This enables the high-level policy to infer optimal execution trajectories and demonstrate zero-shot generalisation when exposed to entirely unseen, complex task structures [Sohn *et al.* 2018]. Although highly effective at traversing variable task strings, sub-task graph solvers are fundamentally bottlenecked by their reliance on discrete, symbolic relationship mappings. These architectures require explicit, pre-defined knowledge of graph structures and crisp logical conditions. Consequently, they lack the flexibility to ingest continuous, high-dimensional sensory data or handle dynamic internal variables, failing completely when task priorities change fluidly as a function of continuous, time-decaying metabolic demands.

3.4.2 State-Bounded Safety Regions in Two-Tier Control

An alternative branch of HRL addresses the problem of physical vulnerability during hierarchical exploration by embedding state-space restrictions directly into the multi-level delegation loop, notably deployed in safety-critical domains like autonomous driving [Gu *et al.* 2023]. This paradigm organises decision-making into a tightly coupled manager-worker architecture. To prevent the low-level worker policy from executing dangerous or unrecoverable actions, the high-level manager policy evaluates the environment's state space to project a mathematically bounded Feasible Safe Region. The high-level manager is restricted to delegating sub-goals that reside strictly within this region. This operational envelope ensures that the low-level worker policy operates exclusively within kinematically reachable and safe trajectories, guaranteeing that the overall system path conforms to a safe global envelope [Gu *et al.* 2023]. While this two-tier safety bounding mitigates immediate collision risks, its defensive constraints are entirely oriented toward external spatial and kinematic properties (such as obstacle coordinates or velocity limits). These frameworks fail to incorporate internal physiological or metabolic states into their hierarchical safety projections. If the high-level safety mask does not monitor time-varying internal resource depletion, the manager may indefinitely restrict the agent to a kinematically "safe" spatial zone where it ultimately terminates due to resource starvation.

3.4.3 Hierarchical RL Literature Gaps

Despite the structural benefits offered by sub-task graphs and safe multi-tier tracking, a profound gap persists at the intersection of temporal abstraction and continuous, multi-resource homeostatic persistence. The existing HRL research landscape displays three primary deficiencies:

- *Decoupling of External Kinematics from Internal Physiological Decay:* State-of-the-art hierarchical safety frameworks, such as Feasible Safe Regions [Gu *et al.* 2023], model safety strictly as a function of external spatial positions or mechanical parameters. They lack the structural mechanisms to ingest continuous, step-by-step internal resource depletion rates (S_P). Because top-tier safety managers are isolated from internal physiological variables, they are incapable of proactively balancing survival needs against external tasks, executing spatially safe paths that still terminate in catastrophic failure (ALIVE $=$ FALSE).
- *Topological Rigidity vs. Combinatorial Option Explosion:* Graph-based sequencing frameworks like NSGS [Sohn *et al.* 2018] achieve generalisation by traversing structured, hard-coded dependency paths. However, when faced with open-ended survival scenarios containing multiple independent resources, the possible combinations of behavioural requirements explode exponentially ($k = 2^{2^n}$). Flat graph solvers suffer from severe scalability limitations under this combinatorial explosion, lacking the algebraic flexibility required to construct complex sub-policies fluidly without manual graph rebuilding.
- *Lack of Integration between Meta-Survival Planning and Low-Level Zero-Shot Engines:* Current HRL literature treats the acquisition of sub-policies and top-level

routing as separate optimisation processes, often requiring extensive low-level pre-training phases. There remains a distinct lack of hierarchical frameworks that pair a high-level value iteration meta-policy—tasked with monitoring physiological resource levels (S_P)—with a low-level compositional engine capable of instantiating target option policies (π_c) zero-shot via Boolean task algebras.

To address these limitations, this dissertation introduces a novel hierarchical compositional reinforcement learning architecture. By explicitly bifurcating the state space into internal survival resource levels (S_P) and external positional coordinates (S_E), the proposed framework breaks the bottleneck of topological rigidity. The top-tier survival policy (π_s) actively tracks multi-resource physiological decay vectors to maintain homeostatic stability, using value iteration to proactively select macro-actions from an exponentially large, combinatorially rich option space ($\Omega = 2^{2n}$). These options are resolved instantaneously at the primitive step level by a zero-shot algebraic compositional engine, satisfying long-horizon environmental objectives without sacrificing optimisation efficiency or absolute survival safety.

3.5 Summary of Literature Gaps

Across the distinct domains of Compositional RL, Safe RL, Survival-Oriented Reinforcement Learning (SORL), and Hierarchical RL, prior literature has established powerful separate mechanisms for skill reuse, safety enforcement, and temporal abstraction. However, when these paradigms are evaluated against the demands of open-ended, multi-resource autonomous survival, their individual and structural limitations become evident. The existing research landscape reveals a critical, interconnected three-part deficiency:

- *The Ingestion Gap (Compositional vs. Physiological):* State-of-the-art compositional and logical task algebras operate under the restrictive assumption of static, external environmental rewards and monolithic state representations. They possess no intrinsic mathematical mechanisms to ingest or prioritise non-linear, step-by-step internal physiological decay functions (S_P).

- *The Horizon Gap (Reactive Safety vs. Proactive Longevity):* Current Safe RL frameworks (such as CMDPs or runtime shielding) are fundamentally reactive, focusing strictly on localised, immediate spatial or kinematic hazard avoidance. Conversely, monolithic survival architectures maximise lifetime from scratch but lack any behavioural modularity, making zero-shot skill reuse impossible.

- *The Architectural Gap (Topological Rigidity vs. Combinatorial Explosion):* Hierarchical sequencing models rely on hard-coded sub-task dependency graphs or discrete knowledge topologies. When exposed to continuous environments with multiple competing resource needs, the resulting target option space explodes combinatorially ($\Omega = 2^{2n}$), inducing optimisation bottlenecks and policy degradation in flat or rigidly structured networks. Ultimately, a profound meta-gap persists: no existing framework unifies multi-resource physiological survival track-

ing, proactive long-term hazard forecasting, and zero-shot algebraic skill composition into a coordinated, multi-tier decision hierarchy.

3.6 Contribution of This Work

This dissertation directly addresses these interconnected literature gaps by introducing the Hierarchical-Compositional Survival Reinforcement Learning (HCSRL) framework. Rather than treating safety, compositionality, and temporal hierarchy as isolated objectives, HCSRL fuses them into a unified, mathematically rigorous architecture. Specifically, this work provides four primary contributions to the field of reinforcement learning:

- *Bifurcated State-Space Formalisation for Homeostatic Tracking:* We introduce a formal state-space decoupling that separates internal physiological resource dynamics (S_P) from external spatial coordinates (S_E). This allows the agent to ingest step-by-step metabolic decay trends directly into its core decision-making loop without destabilising low-level control.
- *Meta-Level Value Iteration over Composed Option Spaces:* We design a high-level survival policy (π_s) that operates over an exponentially large, combinatorially rich option space ($\Omega = 2^{2n}$). By evaluating metabolic deficits at a macro-temporal scale, this meta-policy proactively balances competing survival needs and mitigates the curse of dimensionality inherent in bounded Boolean task algebras.
- *Zero-Shot Low-Level Algebraic Skill Orchestration:* We present a decoupled hierarchical execution loop (formalised in Algorithm 2) where the high-level policy dynamically routes strategies to a low-level, zero-shot compositional engine. This ensures that the agent retains complete behavioural modularity and instantaneous skill reuse across novel environmental tasks without requiring expensive model retraining or fine-tuning.
- *Empirical Validation of Proactive Survival Persistence:* Through extensive simulation tracking across high-risk, resource-scarce domains, we demonstrate that the proposed architecture prevents catastrophic terminal threshold breaches (ALIVE $=$ FALSE) and significantly extends continuous life expectancy, outperforming both traditional flat constraint-satisfaction agents and model-free hindsight architectures.

Chapter 4

Hierarchical Compositional Survival-Oriented RL

The proposed Hierarchical-Compositional Survival-Oriented Reinforcement Learning (HCSRL) framework operationalises survival as a multi-level decision-making problem. Designed to navigate the intrinsic trade-off between resource acquisition and hazard avoidance, the architecture leverages the zero-shot properties of Boolean Task Algebra within a temporal hierarchy [Tanık and Ertekin 2024]. This allows the agent to dynamically adapt to fluctuating physiological needs without the prohibitive computational overhead of retraining for every environmental contingency.

The fundamental departure from traditional Hierarchical Reinforcement Learning (HRL), such as the Options framework, lies in the definition of the objective function. While standard HRL typically optimises for reaching a discrete terminal goal state, HCSRL treats survival as an emergent property of correctly composed behaviours. Instead of simple goal-attainment, the framework focuses on trajectory maintenance i.e. optimising a path through the bifurcated state space that maximises life expectancy. By applying algebraic logic to a library of base skills, the agent can compose complex survival strategies (e.g., *Seek Water* $\wedge$ $\neg$*Avoid Hazard*) in real-time, ensuring that physiological variables remain within viable homeostatic bounds.

4.1 Formalising Base Skills via Goal-Conditioned MDPs

To circumvent the prohibitive computational overhead of retraining for every emergent combination of physiological needs, we establish a modular library of reusable survival base tasks $\mathcal{T}$. Each survival-critical skill t e.g., *Resource Acquisition* or *Hazard Avoidance* is formalised as a Goal-Conditioned Markov Decision Process (GCMDP), defined by the tuple $\langle \mathcal{S}^{\mathcal{E}}, \mathcal{A}, P, \mathcal{G}, R, \gamma \rangle$. Within this framework, the environmental state space $\mathcal{S}^{\mathcal{E}}$ characterises the agent's external context, such as its position, local sensor observations, and relative proximity to environmental landmarks. Actions $a \in \mathcal{A}$ represent the fundamental motor primitives available to the agent, providing the necessary granularity for executing complex behaviours. We assume deterministic transition dynamics

$P(s^{e'} \mid s^e, a)$, where the evolution of the environmental state is uniquely determined by the execution of the goal-conditioned policy $\pi(a \mid s^e, g)$. The goal space $\mathcal{G} \subset \mathcal{S}^{\mathcal{E}}$ represents the subset of states corresponding to successful task completion, for instance, the spatial coordinates where a resource is accessible. To facilitate robust skill acquisition, the Worker agent compiling this library, utilises a sparse internal reward function R. This signal prioritises efficiency, incentivising the agent to identify the optimal trajectory towards a target or the most rapid withdrawal from a hazardous region.

4.2 The Compositional Worker: Zero-Shot Execution

The fundamental challenge in survival RL is the "combinatorial explosion" of states; an agent might be hungry, thirsty, and near a hazard simultaneously. Standard RL would require training a unique policy for every permutation of these states. To overcome this, we utilise *Extended Value Functions (EVFs)* [Nangue Tasse *et al.* 2020]. This design decision allows the Worker to derive the optimal policy for any complex goal $c_i \in \Omega$ issued by the Manager without additional training or environment samples. By treating pre-trained base Q-values as a basis set, the Worker applies algebraic operators $(\neg, \vee, \wedge)$ directly to the value functions. For example, when the Manager identifies a state of being "Thirsty but near a Hazard," it does not need to search for a specific "Thirsty-Hazard" policy. Instead, the Worker instantly composes the Q_water $\wedge \neg Q$_hazard (see Equations 2.14 and 2.15). The resulting policy $\pi_{\text{water}\wedge\neg\text{hazard}}$ is executed zero-shot, providing immediate adaptation to life-threatening scenarios that were never explicitly encountered during the initial training of the primitive skills.

The Worker executes a goal-conditioned policy π_{c_i} over a variable time horizon τ. During each discrete step of this interval, the agent selects a primitive action $a \in \mathcal{A}$ based on the current environmental state s^e to navigate toward the manifold $\mathcal{G}$. A critical feature of this loop is the continuous coupling of the bifurcated state space; every environmental transition incurs a metabolic cost, reflected by a decrement in the physiological state $s^{p'} = s^p - 1$. This mechanism ensures that spatial navigation is intrinsically bound to energy expenditure, transforming time into a finite resource that the agent must manage to maintain homeostasis. Survival is operationalised through strict terminal conditions and a specialised reward structure. If the resource levels in $\mathcal{S}^{\mathcal{P}}$ are depleted or reach toxic levels, the agent enters a non-ALIVE state, triggering a catastrophic penalty that is back-propagated to the high-level controller. Conversely, successful arrival at a goal location sets the DONE flag to true and replenishes s^p based on the resources available at that coordinate. By accumulating these step-wise rewards and penalties into a cumulative signal r_c, the framework enables the Manager to treat survival not as a static goal, but as an emergent property of a correctly maintained trajectory through both the physiological and environmental manifolds.

4.2.1 Methodological Justification for Tabular Implementations

While tabular value-representation scales poorly to high-dimensional or continuous environments, it is deliberately utilised here to establish a clean mathematical baseline for evaluating the HCSRL framework. By guaranteeing exact convergence without

the confounding optimisation errors common to deep function approximation—such as catastrophic forgetting or gradient instability—tabular Q-learning isolates the algebraic operations of Boolean Task Algebra. This ensures that the empirical findings strictly evaluate the efficacy of the hierarchical survival logic rather than the idiosyncratic stability of neural network training vectors. Transitioning these primitives to deep architectures (such as DQN or PPO models) remains an architectural extension detailed in Section 7.3.

4.3 The Survival Manager: High-Level Task Selection

At the top of the hierarchy, the *Survival Manager* (π_s) operates as a meta-policy. Its objective is to monitor the state $\mathcal{S}$ and determine which combination of tasks will best maximise life expectancy. Unlike traditional Hierarchical Reinforcement Learning (HRL), where the manager selects from a small set of discrete, manually defined skills, the HCSRL manager selects from the *Boolean Task Space* Ω.

To capture the temporal persistence required for survival, such as the time it takes to travel to a water source once thirsty, we formulate the Manager's decision-making process as a Semi-Markov Decision Process (SMDP) [Howard 1971]. The Manager operates at a higher level of abstraction, selecting compositions of primitive skills that persist over a variable number of time steps. We define the Manager's SMDP as the tuple $\langle \mathcal{S}, \Omega, P, R_c, \gamma \rangle$. The bifurcated state space $\mathcal{S} = [\mathcal{S}^{\mathcal{P}}, \mathcal{S}^{\mathcal{E}}]$ decouples internal physiological imperatives from external spatial navigation. The physiological state ($\mathcal{S}^{\mathcal{P}}$) is a vector representing internal resource levels (e.g., energy, hydration). These values are dynamic and decay over time, effectively defining the agent's "urgency". The environmental state ($\mathcal{S}^{\mathcal{E}}$) is the agent's external context, such as coordinates (x, y) in a grid or raw pixel values, used to localise resources and hazards in the world. The action space (Ω) is the set of possible *algebraic compositions* of worker skills. Instead of selecting a primitive action $a \in \mathcal{A}$, the Manager selects a composition $\omega \in \Omega$ (e.g., Seek(Water) $\wedge \neg$Enter(Hazard)). Given the hierarchical nature, the transition $P(s'|s, \omega)$ is stochastically determined by the underlying Worker's policy π_w. If the Manager selects composition ω at state s, the Worker executes the corresponding composed policy π_ω for τ steps until a termination condition is met (e.g., the physiological need is saturated or a fixed time horizon is reached). The next state s' is then drawn from the multi-step transition: probability

$$P(s', \tau|s, \omega) = \prod_{t=1}^{\tau} P_{env}(s_{t+1}|s_t, \pi_\omega(s_t))$$

The feedback loop between the hierarchical levels is closed by the cumulative reward r_c. Rather than a static success bonus, r_c represents the temporal integration of the Worker's step-wise rewards and penalties over the interval τ. This signal incorporates a marginal step bias ($+0.001$) to encourage rapid goal attainment, while simultaneously embedding the heavy survival penalty inherited from the Worker node upon physiological depletion. By back-propagating this aggregated value into the High-Level Q-

function, the Manager learns to compose tasks that are not only effective in reaching environmental goals but are also metabolically sustainable.

4.3.1 Formalisation of the Hierarchical Functional Split

To prevent policy-capability confusion and ensure structural modularity, the proposed framework enforces a strict, non-overlapping functional split between the high-level Survival Manager and the low-level Worker. This operational bifurcation decouples metabolic urgency from spatial navigation, mapping their distinct characteristics across state spaces, action spaces, temporal horizons, and optimisation objectives.

The precise boundary governing this hierarchical split is systematically defined across the following four dimensions:

1. **State-Space Partitioning ($\mathcal{S}^{\mathcal{P}}$ vs. $\mathcal{S}^{\mathcal{E}}$):** The Manager possesses no direct awareness of the agent's exact spatial coordinates within the grid world; it operates exclusively over the physiological state space $\mathcal{S}^{\mathcal{P}}$ (monitoring internal hydration, nutrition, and energy meters). Conversely, the Worker is blind to the agent's internal biological urgency, operating strictly within the external spatial coordinate state space $\mathcal{S}^{\mathcal{E}}$.

2. **Action-Space Separation ($\mathcal{G}$ vs. $\mathcal{A}$):** The Manager's action space consists of high-level goal configurations $\mathcal{G}$, selected through algebraic task composition (e.g., instructing the agent to seek food, seek water, or satisfy both simultaneously). The Worker executes no goal-setting logic; its action space is restricted to immediate, reactive environmental movement primitives $\mathcal{A} = \{\text{North}, \text{South}, \text{East}, \text{West}\}$.

3. **Temporal Abstraction (Macro vs. Micro Scale):** The Manager operates on a semi-MDP temporal scale, emitting a high-level logical objective that remains active over a multi-step macro horizon τ. The Worker processes operations at a fine-grained, step-by-step micro scale (t), executing continuous positional changes until the assigned spatial target is reached or the macro horizon expires.

4. **Optimisation Objectives (Homeostasis vs. Navigation):** The Manager is optimised solely to maximise overall life expectancy and maintain internal resource levels within safe homeostatic envelopes. The Worker is optimised purely for path efficiency, training via Extended Value Functions (EVFs) to find the shortest trajectory to whatever spatial distribution satisfies the Manager's current logical instruction.

To ensure absolute architectural clarity, the precise parameters bounding these two entities are summarised in Table 4.1.

4.4 Summary of the HCSRL

1. **Perception:** Observe the current bifurcated state $\mathcal{S} = [\mathcal{S}^{\mathcal{P}}, \mathcal{S}^{\mathcal{E}}]$, where $\mathcal{S}^{\mathcal{P}}$ denotes internal physiological needs (e.g., hunger, thirst) and $\mathcal{S}^{\mathcal{E}}$ denotes the external environmental context (e.g., ego-position, sensor data).

Table 4.1: Operational and Mathematical Delineation of the Manager-Worker Hierarchy.

Architectural Attribute	High-Level Survival Manager	Low-Level Base Worker
Operational Scope	Internal Physiology & Urgency	External Spatial Navigation
State Space Input	$\mathcal{S}^{\mathcal{P}} = [h_t, n_t, e_t]^T$	$\mathcal{S}^{\mathcal{E}} = [x_t, y_t]^T$
Action Space Output	Composed Boolean Goals $\mathcal{G}$	Grid Movement Primitives $\mathcal{A}$
Temporal Horizon	Macro (Variable Multi-Step, τ)	Micro (Atomic Step-wise, t)
Primary Objective	Multi-Resource Homeostasis	Shortest-Path Local Tracking
Underlying Calculus	Boolean Task Algebra Engine	Extended Value Functions (EVFs)

2. **Composition:** The Manager selects a complex logical task $c_i \in \Omega$ by arbitrating between competing survival drives, represented as an algebraic combination of base skills (e.g., $t_1 \wedge \neg t_2$).
3. **Synthesis:** The Worker derives the optimal goal-conditioned policy π_{c_i} zero-shot by applying Boolean Task Algebra to the library of pre-trained Extended Value Functions (EVFs).
4. **Action:** The Worker executes primitive control actions $a \in \mathcal{A}$ within a Semi-Markov Decision Process (SMDP) framework. This loop persists until the composition goal is reached (DONE) or the agent suffers physiological depletion ($\neg$ALIVE).
5. **Update:** The High-level Q-values are updated based on the cumulative reward r_c, which integrates the sum of marginal step biases and any catastrophic survival penalties incurred during the Worker's trajectory.

4.5 Motivation: The Architecture of HCSRL

The design of the HCSRL framework is motivated by the fundamental discrepancy between how standard Reinforcement Learning (RL) defines "tasks" and how biological systems navigate "survival" [Jiang *et al.* 2024]. Standard RL typically operates on a goal-attainment paradigm, where an agent's success is binary and tied to a specific state. In contrast, survival is an unending, multi-objective, and temporally extended trajectory maintenance problem. The following design decisions motivate our specific hierarchical approach.

4.5.1 Bifurcation of State: Decoupling Need from Capability

A primary motivation for splitting the state space into Physiological ($\mathcal{S}^{\mathcal{P}}$) and Environmental ($\mathcal{S}^{\mathcal{E}}$) manifolds is to solve the utility-capability confusion found in "flat" RL agents. In a flat architecture, an agent's policy for "seeking water" might change based on how thirsty it is. This is computationally inefficient. By restricting the Worker to $\mathcal{S}^{\mathcal{E}}$, we ensure that the physical skill—navigating to a coordinate—is a stable, reusable primitive. The Manager alone handles $\mathcal{S}^{\mathcal{P}}$, determining the *urgency* of that skill. This

decoupling allows the agent to maintain high-performance navigation regardless of its internal metabolic "panic."

4.5.2 Boolean Task Algebra: Solving the Combinatorial Explosion

In safety-critical survival, the environment is rarely "clean." An agent rarely just needs "Food"; it needs "Food AND NOT Hazard" or "Water OR Food AND NOT Hazard." If we used standard RL for every combination, the number of required policies would grow exponentially with the number of resources and hazards (2^n). We motivate the use of Boolean Task Algebra to enable Zero-Shot Composition. By using EVFs, the agent treats learned skills as a "basis set." This allows the agent to handle novel, complex survival threats—such as a hazard appearing on the only path to a vital resource—without a single step of additional gradient descent.

4.5.3 SMDP Formalism: Temporal Persistence and Metabolic Costs

Survival requires a commitment to a course of action over time. A "per-step" decision-making process at the high level often leads to "jitter," where an agent oscillates between two needs (e.g., moving one step toward food, then one step toward water) and satisfies neither. Formulating the Manager as a Semi-MDP allows it to "lease" control to the Worker for a duration τ. This ensures the agent follows a consistent trajectory until a physiological need is met or a critical threat is avoided. This architecture explicitly motivates survival by tying the passage of time (τ) to a decay in $\mathcal{S}^{\mathcal{P}}$. Because every step taken by the Worker costs energy, the Manager is naturally incentivised to select the most "logical" algebraic composition to minimise time-to-goal and maximise life expectancy.

4.5.4 Trajectory Maintenance over Goal Attainment

The HCSRL framework is motivated by the shift from terminal rewards to trajectory alignment. In survival, there is no "winning" state, only the avoidance of "losing" states (death). By integrating the Worker's performance over the SMDP interval τ, the Manager receives a signal that represents the "healthiness" of a behavioural mode. This allows the framework to align agent behaviour with long-term survival rather than short-term reward hacking, ensuring that the agent doesn't just "reach the goal" but does so in a way that leaves it in a viable state to pursue the next requirement.

4.6 Running Example

To illustrate the HCSRL framework, we consider a 20×20 grid world environment containing an agent and multiple resource cells (Food and Water), see Figure 4.1. This example instantiates the components defined in Algorithm 2. The environment state is partitioned into physiological state $\mathcal{S}^{\mathcal{P}} = [f, w]$, where f (food) and w (water) represent the agent's internal survival resources randomly generated between 0 and 7, and positional state $\mathcal{S}^{\mathcal{E}} = [x, y]$, representing the agent's Cartesian coordinates on the

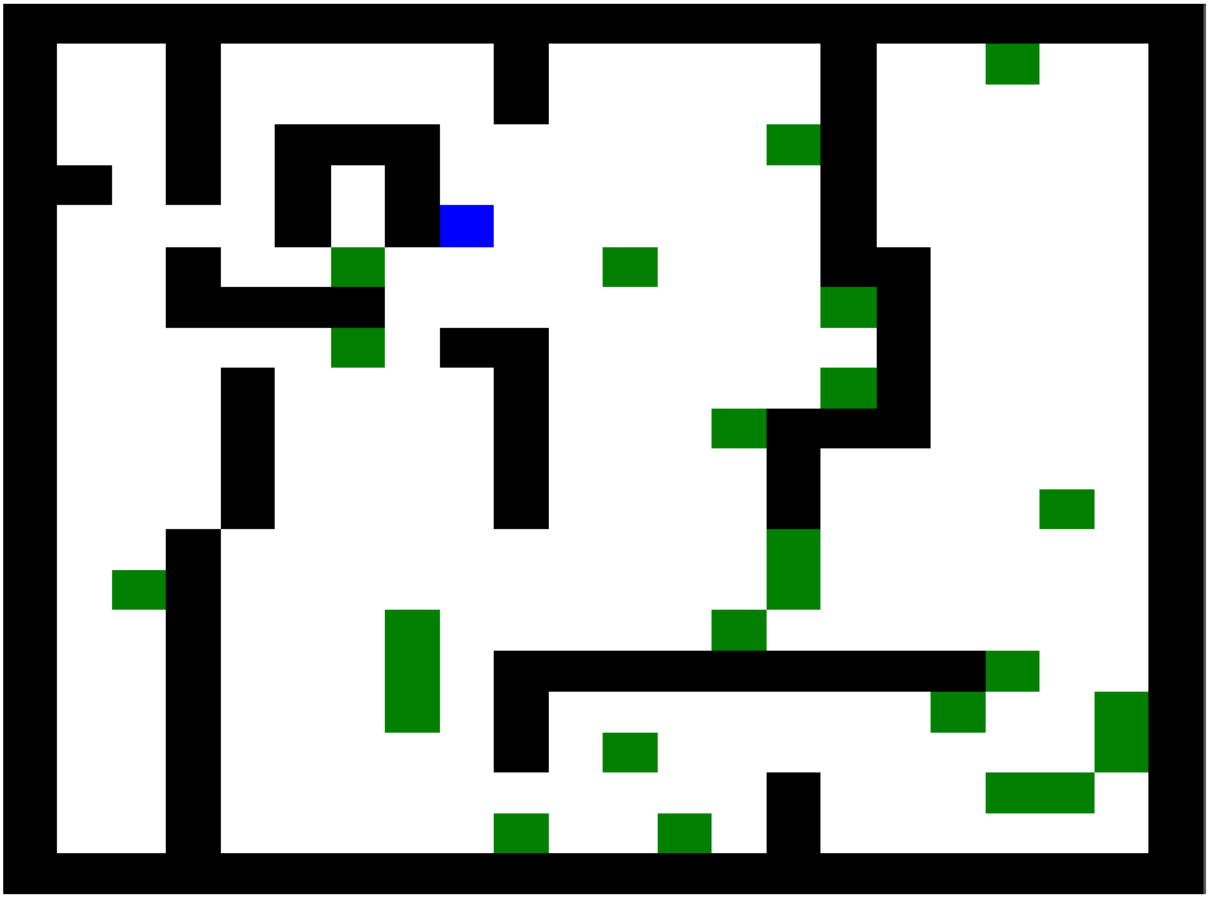


Figure 4.1: This figure illustrates the HCSRL framework within a 20 $\times$ 20 grid world containing an agent and multiple resource cells. The environment instantiates the components of Algorithm 2, where the state space S ($\approx$ 30 000 states) is defined by the Cartesian product of the positional manifold $\mathcal{S}^{\mathcal{E}} = [x, y]$ and the physiological manifold $\mathcal{S}^{\mathcal{P}} = [f, w]$. Here, f (food) and w (water) represent the agent's internal survival resources, randomly generated between 0 and 7. The agent interacts with the grid using a primitive action set A=Up, Down, Left, Right to complete base tasks $\mathcal{T} = \{t_{\text{food}}, t_{\text{water}}\}$.

20×20 grid. The state space $\mathcal{S}$ is a finite discrete set defined by the Cartesian product of the environmental and physiological manifolds, $\mathcal{S} = \mathcal{S}^{\mathcal{E}} \times \mathcal{S}^{\mathcal{P}}$. The environment has approximately 30,000 states and on the order of 10^4 states. The set of primitive actions is defined as the finite set $\mathcal{A} = \{\text{Up, Down, Left, Right}\}$. We establish two base tasks $\mathcal{T} = \{t_{\text{food}}, t_{\text{water}}\}$, where each task corresponds to a policy trained to reach its respective resource cell. Following the logic of Boolean Task Algebra, this basis set yields $k = 2^{2^{|\mathcal{T}|}} = 16$ possible compositional options Ω available for selection by the Manager.

In alignment with the low-level execution loop of the framework, the environment enforces a continuous metabolic cost through Resource Decay, where the physiological state s^p is decremented at every discrete time step such that $f_{t+1} = f_t - 1$ and $w_{t+1} = w_t - 1$. This decay directly informs the agent's Survival Limits (ALIVE); the agent remains operational only while resource levels stay within strictly defined homeostatic bounds, specifically $0 < f < 20$ and $0 < w < 21$. If these resources reach depletion (0) or exceed their upper bounds through over-consumption, the ALIVE flag is set to FALSE, the SMDP loop terminates immediately, and the Manager receives a catastrophic penalty of -1. Conversely, Task Completion (DONE) occurs when the positional state s^e matches a goal cell, triggering a replenishment of resource levels and a successful return of control to the High-Level policy. This dynamic ensures that survival is an emergent property of maintaining a viable trajectory within a finite, bifurcated state space.

The reward structure r_c used for training the Manager is an integrated signal accumulated during the low-level execution phase. It consists of a step reward, providing a marginal positive bias of $+0.001$ per step to incentivise temporal efficiency and goal-oriented movement, alongside a survival penalty, which imposes a terminal reward of -1 if the agent fails to maintain its physiological state within the designated survival bounds. By aggregating these signals, the framework ensures that the Manager optimises for trajectories that are both efficient and biologically sustainable.

Following the High-Level policy (π_s) outlined in Algorithm 2, the Manager performs an observation of the full bifurcated state $\mathcal{S} = [f, w, x, y]$. If the internal resource f is dangerously low while w remains sufficient, the Manager selects the compositional option $c = t_{food}$. Conversely, if both resources are moderately depleted, the Manager may select the logical conjunction $c = t_{food} \wedge t_{water}$. The Worker then synthesises the policy for this specific composition zero-shot via Boolean Task Algebra, navigating to the environmental coordinates most likely to extend the agent's ALIVE duration. This decision process ensures that the agent's external trajectory is always a direct function of its internal homeostatic requirements.

To account for the temporal costs of these multi-step trajectories, the framework incorporates a discounting factor γ. Within the Semi-MDP structure, the High-Level Manager updates its value function based on the discounted cumulative reward r_c collected by the Worker over the variable duration τ. Specifically, the future value is discounted by γ^τ, where τ represents the total number of primitive steps taken until the task is marked as DONE or the agent is no longer ALIVE. This discounting mechanism ensures the Manager prioritises compositions that achieve physiological replenishment quickly, as the utility of resources diminishes the longer the agent remains in a state of depletion. This decision process ensures that the agent's external trajectory is always a

direct function of its internal homeostatic requirements, balanced against the temporal urgency of survival.

Algorithm 2 modelling Survival through the composition of extended Q-value functions

Input :

- Environment $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma)$
- State space $\mathcal{S} = [\mathcal{S}^{\mathcal{P}}, \mathcal{S}^{\mathcal{E}}]$ where $\mathcal{S}^{\mathcal{P}}$ is the survival resources levels of the agent and $\mathcal{S}^{\mathcal{E}}$ is the positional state of the agent, i.e. grid or pixel position
- Set of primitive actions $\mathcal{A} = \leftarrow, \rightarrow, \uparrow, \downarrow$
- Set of base tasks $\mathcal{T} = \{t_1, t_2, ...t_n\}$ that are already trained to reach goal positions that contain a specific resource, where each $t_i = (I_i, \pi_{t_i}, \beta_i)$
- All possible compositional RL options $\Omega = c_1, c_2, ..., c_k$ where $k = 2^{2^n}$
- Survival High-level policy π_s
- Zero-shot algorithm for Low-level policies (option policies), i.e $\pi_{c_i}\ \forall\ c_i \in \Omega$
- Learning rate α, discount factor γ and exploration constant ϵ

Initialise : Initialize parameters for π_s and all π_ω

for $e = 1, 2, ..., x$ **do**

Set initial survival state $s_c = [s^p, s^e]$

ALIVE = TRUE

Zero-shot compositional RL policies π_{c_i} for all c_i

while survival agent is ALIVE **do** ▷ Survival High-level policy

Select compositional RL options $c_i = \begin{cases} \arg\max_{c_i} \bar{Q}^*_{c_i}(s_c, c_i) & \text{with probability } 1-\epsilon \\ \text{any } c_i & \text{with probability } \epsilon \end{cases}$

DONE = FALSE

$r_c = 0$

while Not DONE or ALIVE **do** ▷ Low-Level Execution (Option)

Select a primitive action a from $\mathcal{A}$ using $\pi_{c_i}(s^e)$

Perform action a, observe step reward r and next state $s^{e\prime}$

$s^{p\prime} = s^p - 1$

$r_c = r_c + r$

if composition goals reached **then**

DONE = TRUE

$s^{p\prime} = s^p +$ resources in that goal location

if in state $s^{e\prime}$ and there are not enough survival resources $s^{p\prime}$ **then**

ALIVE = FALSE

$r_c = r_c + \mathbf{penalty}$

if still Not DONE and ALIVE **then**

$s^e = s^{e\prime}$

Observe composition reward r_c, and next survival state $s_c{}' = [s^{p\prime}, s^{e\prime}]$

$Q(s_c, c) \leftarrow (1-\alpha) \cdot Q(s_c, c) + \alpha \cdot (r_c + \gamma \cdot \max_{c'} Q(s_c{}', c'))$

$s_c = s_c{}'$

Chapter 5

Experiments

This chapter details the empirical evaluation of the *Hierarchical-Compositional Survival Reinforcement Learning (HCSRL)* framework. While the previous chapter established the theoretical viability of using Boolean Task Algebra for survival, this section aims to validate its performance against established benchmarks: *Standard Q-Learning* and *Goal-Oriented Q-Learning GORL*. The primary objective is to determine how effectively the HCSRL approach enables an agent to navigate a dynamic environment where resource scarcity, temporal decay, and physiological limits impose constant termination pressure. To ensure a rigorous and unbiased comparison, all methodologies are benchmarked within a unified grid-world survival environment. In this setting, the agent is tasked with maintaining two internal state variables—*food* and *water*—which decay at fixed rates. This creates a "persistent survival envelope" wherein the agent must proactively balance resource acquisition; failure to do so results in termination due to either deprivation (depletion) or toxicity (excess). This environment serves as an ideal stress test for long-horizon decision-making and zero-shot skill composition. Our investigation is guided by the following key research questions:

- *Persistence and Policy Robustness:* To what degree does the HCSRL approach extend the mean life expectancy of the agent compared to monolithic and non-compositional baselines?
- *Modular Sample Efficiency:* Does the decomposition of survival into base skills (food-seeking vs. water-seeking) accelerate the convergence of the high-level policy?
- *Operational Complexity:* What is the relative computational overhead of each method in terms of training duration, memory footprint, and policy execution?

To systematically address these research questions, each algorithm is trained under a strictly identical computational budget of *15,000 episodes*. Performance is quantified through a multi-dimensional evaluation framework: we measure *cumulative episode steps* to track life expectancy, analyse *convergence rates* to assess learning speed, and verify *policy stability* across multiple stochastic environmental seeds. Furthermore, to provide a statistically rigorous assessment of agent persistence, we employ *Kaplan–Meier survival* analysis, a non-parametric statistic used to estimate the survival function from

life-duration data [Kaplan and Meier 1958]. The subsequent sections detail the specific environmental parameters, hyper-parameter configurations, and the statistical procedures utilised to interpret these results.

5.1 Environment Configuration and Parameters

The survival grid world is designed to simulate a resource-constrained ecosystem. The environment is implemented as a 20×20 discrete grid where the agent's survival is a function of spatial navigation and internal state management, see Table 5.1.

Parameter	*Value*	*Description*
Food Capacity (f_{max})	20 units	Upper survival limit for the food resource.
Water Capacity (w_{max})	21 units	Upper survival limit for the water resource.
Decay Rate (Δs)	-1 unit/step	Constant metabolic drain on both f and w at each time step.
Goal Replenishment	Variable	Quantity of resources acquired upon reaching a goal cell. Food and water resources at each goal are randomly generated between 0 and 7
Initial State	Fixed	Agent starts at coordinates $(10, 10)$ with $f = 20$ and $w = 20$.
Experiment Seeds	Random	30 independent random seeds used for statistical validation.

Table 5.1: Environmental Configurations and Parameters

To ensure the comparison between HCSRL, Standard Q-Learning, and Goal-Oriented Q-Learning is valid, we maintain consistent learning parameters across all architectures where applicable, see Table 5.2.

While the discrete, static 20×20 grid world lacks the environmental noise of physical deployments, it provides a strictly controlled benchmark necessary for isolating the algebraic validity of the HCSRL framework. By minimising confounding variables such as function approximation errors, erratic sensor inputs, or unpredictable transition dynamics, this environment allows for a direct, mathematically transparent evaluation of how high-level logical compositions affect multi-resource homeostatic stability. Consequently, this setup serves as a foundational baseline before scaling the framework to continuous or physically embodied domains.

5.1.1 Characterisation of Environmental Dynamics and Uncertainty Boundaries

It is important to explicitly define the nature of the environmental dynamics evaluated in these experiments. The dynamic parameters of the simulation are strictly internal

Hyper-parameter	*Symbol*	*Value*
Learning Rate	α	0.001
Discount Factor	γ	0.99
Exploration Strategy	ϵ-greedy	Starts at 1.0, exponentially decays to 0.05 over $15,000$ episodes.
Survival Penalty	ζ	-1.0 (applied upon ALIVE = FALSE).
Step Reward	r	$+0.001$ (to encourage movement).
Total Episodes	E	$15,000$

Table 5.2: Hyper-parameter Configurations

and physiological, governed by the continuous step-wise decay of the agent's metabolic variables (hydration, nutrition, and energy). External environmental variables, such as resource coordinates and spatial transition dynamics, are kept deterministic and static upon episode initialisation.

This experimental setup intentionally omits sensory noise, partial state observability, or moving hazards to maintain a rigorous scientific control. Isolating the framework from external stochastic variables ensures that the empirical results strictly evaluate the mathematical stability and behavioural accuracy of the algebraic skill compositions under homeostatic stress. Transitioning these structural primitives to partially observable or non-stationary environments is formalised as an architectural extension in Section 7.3.

5.2 Statistical Procedures and Survival Analysis

The evaluation of persistence in Reinforcement Learning requires moving beyond simple mean rewards/steps. This study utilises two primary statistical methods to interpret agent performance:

5.2.1 Kaplan-Meier Survival Analysis

We treat the agent's life duration (total steps per episode) as "time-to-event" data. The Kaplan-Meier estimator is used to calculate the survival function $S(t)$, representing the probability that an agent remains "ALIVE" past a certain time step t:

$$S(t) = P(T > t)$$

This allows us to visualise the "mortality" of different policies. If the HCSRL curve stays consistently above the baseline curves, it provides statistically significant evidence of superior proactive survival.

5.2.2 Convergence and Stability Analysis

Learning curves are smoothed using a moving average window of 100 episodes to filter stochastic noise. We evaluate Policy Stability by calculating the variance in life expectancy across the final 1,000 episodes of training, ensuring that the learned survival strategy is robust and not a result of "lucky" state-space trajectories.

5.3 Baseline Descriptions

To validate the efficacy of the Hierarchical-Compositional Survival Reinforcement Learning (HCSRL) framework, we benchmark its performance against two distinct levels of reinforcement learning complexity. These baselines provide a comparative ladder, moving from a structureless "flat" approach to a non-compositional hierarchical approach.

5.3.1 Benchmark 1: Standard Q-Learning (Flat RL)

The Standard Q-Learning baseline represents a traditional end-to-end learning approach where the agent must discover survival strategies without prior knowledge or task-specific structure.

Key Characteristics:

- *Action Space:* The agent is restricted to primitive moves: $\mathcal{A} = \{\text{UP, RIGHT, LEFT, STAY}\}$.
- *Interaction:* The agent interacts directly with the environment, updating Q-values based on the immediate survival feedback.
- *Policy Mapping:* The learned policy $\pi(s)$ maps each state directly to one of the four primitive actions to maximise long-term survival.

Objective: To learn a complete survival policy from scratch, requiring the agent to implicitly associate specific grid coordinates with resource replenishment and metabolic decay.

5.3.2 Benchmark 2: Goal-Oriented Q-Learning

This benchmark introduces a hierarchical structure but lacks the algebraic compositionality of the HCSRL framework. It operates through a Two-Stage Learning process, where the agent first masters reaching individual goals before learning how to sequence them for survival.

Two-Stage Learning Process:

1. *Pre-training Stage:* The agent learns n optimal policies for every individual goal in the environment (e.g., π_{goal_A} for reaching goal A, which might have food, π_{goal_B} for reaching goal B, which might have food and water).
2. *Survival Learning Stage:* Q-Learning is applied at a higher level of abstraction. In this stage:

- *Actions* are no longer primitive moves but are the pre-learned goal-specific policies.
- *Action Space:* $\mathcal{A}_{\text{high}} = \{\pi_{\text{goal}_1}, \pi_{\text{goal}_2}, \ldots, \pi_{\text{goal}_n}\}$.

Objective: The agent seeks to survive by selecting which goal-oriented policy to follow at each state. While this improves efficiency over the flat baseline, it is fundamentally limited by its inability to compose goals; the agent must choose between seeking food *or* seeking water, whereas the HCSRL agent can pursue both goals via logical conjunction.

5.4 Summary of Baseline Comparison

The following table highlights the architectural constraints of the benchmarks relative to our proposed framework, see Table 5.3:

Feature	*Standard Q-Learning*	*Goal-Oriented Q-Learning*	*HCSRL (Ours)*
Learning Type	End-to-End	Two-Stage (Hierarchical)	Two-Stage (Hierarchical)
Action Space	Primitive Moves	Goal-Oriented Policies	*Algebraic Compositions*
Logical Synthesis	None	Selection Only	*Full Boolean Algebra*
Zero-Shot	No	No	*Yes (2^{2^n} options)*

Table 5.3: Architectural Comparison of Standard Q-Learning, Goal-Oriented Q-Learning, and HCSRL.

Chapter 6

Results and Analysis

The primary objective of this thesis was to develop a hierarchical framework capable of managing complex survival tasks through Boolean logic. This chapter presents the empirical results of the experiments outlined in the previous chapter, providing a comparative analysis of the Hierarchical-Compositional Survival Reinforcement Learning (HCSRL) framework against Standard Q-Learning and Goal-Oriented Q-Learning (GORL) baselines.

The evaluation focuses on how these different architectures perform as the environment scales in complexity—specifically, as the density of resource goals increases. By examining survival outcomes across a spectrum of 10 to 100 goals, we identify the operational boundaries of traditional RL methods and demonstrate the scalability of compositional logic.

The analysis is structured around the following key performance dimensions:

1. *Survival Persistence:* Evaluated using mean life expectancy and Kaplan-Meier survival curves to determine the probability of agent persistence over time.

2. *Statistical Significance:* Utilising Log-Rank tests to provide a rigorous mathematical basis for the performance differences observed between the models.

3. *The Complexity Threshold:* An investigation into the "crossover point" where the benefits of algebraic composition outweigh the simplicity of goal-selection.

4. *Learning Efficiency:* A comparison of convergence rates and policy stability across varying environmental seeds.

By synthesising these results, we aim to demonstrate that while primitive and goal-oriented methods are sufficient for simple tasks, the ability to compose tasks algebraically is a prerequisite for robust survival in dense, multi-objective environments.

6.1 Learning Performance and Convergence

In evaluating the reinforcement learning performance and convergence results, we observed significant differences among the various architectures employed. The analysis focuses on key metrics such as convergence rates, policy stability, and overall learning efficiency. By comparing the Hierarchical-Compositional Survival Reinforcement Learning (HCSRL) framework with the Goal-Oriented Q-Learning (GORL) and Standard Q-Learning baselines, we can discern how each method adapts to complex survival tasks and scales with increasing goal density. These insights reveal the strengths and limitations of each approach, ultimately highlighting the advantages of a hierarchical compositional strategy in dynamic environments.

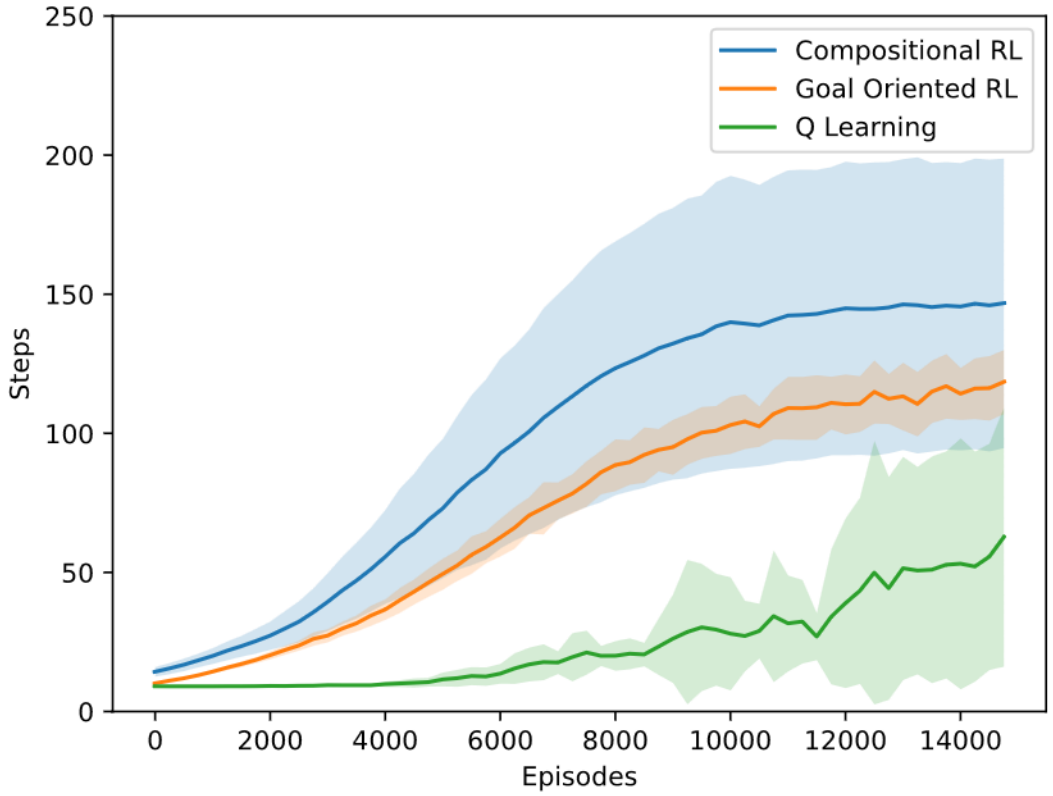


Figure 6.1: The results highlight a trade-off between performance and stability, with Compositional RL achieving the highest average steps but greater variance, Goal-Oriented RL showing steadier moderate gains, and Q-Learning progressing more slowly with lower efficiency.

6.1.1 Training Curves: Average Steps per Episode

The learning curves illustrated in Figure 6.1 represent the mean survival time (measured in steps) across all experimental configurations over 15,000 training episodes.

These curves reveal distinct differences in how each architecture navigates the survival state-space.

1. The HCSRL framework (blue curve) demonstrates superior learning throughout the training duration. Unlike the baselines, HCSRL exhibits a steep upward trajectory within the first 4,000 episodes, showing rapid initial convergence. This is primarily due to the Two-Stage Learning approach, where pre-trained base skills for food and water acquisition allow the high-level manager to focus solely on the logic of survival rather than primitive navigation. By episode 10,000, HCSRL reaches a plateau of approximately 150 steps per episode. While the shaded region indicates variance across seeds, the mean consistently remains higher than both GORL and Q-learning, suggesting a more robust survival policy.
2. The GORL baseline (orange curve) follows a similar learning trend but settles at a significantly lower ceiling of roughly 110–120 steps. While GORL benefits from pre-trained skills, it lacks the algebraic ability to combine them. The learning curve reflects the performance cost of "oscillating" between single goals—the agent learns to survive by alternating needs, but it cannot optimise trajectories by pursuing multiple needs simultaneously, leading to earlier termination.
3. The Q-learning baseline (green curve) exhibits the poorest performance and highest instability. Significant survival progress only begins after 8,000 episodes. The wide shaded area indicates extreme sensitivity to initial conditions (seeds). Without a hierarchical structure, the agent must learn the correlation between positional coordinates and internal physiological decay from scratch, a task that proves inefficient within the 15,000-episode budget.

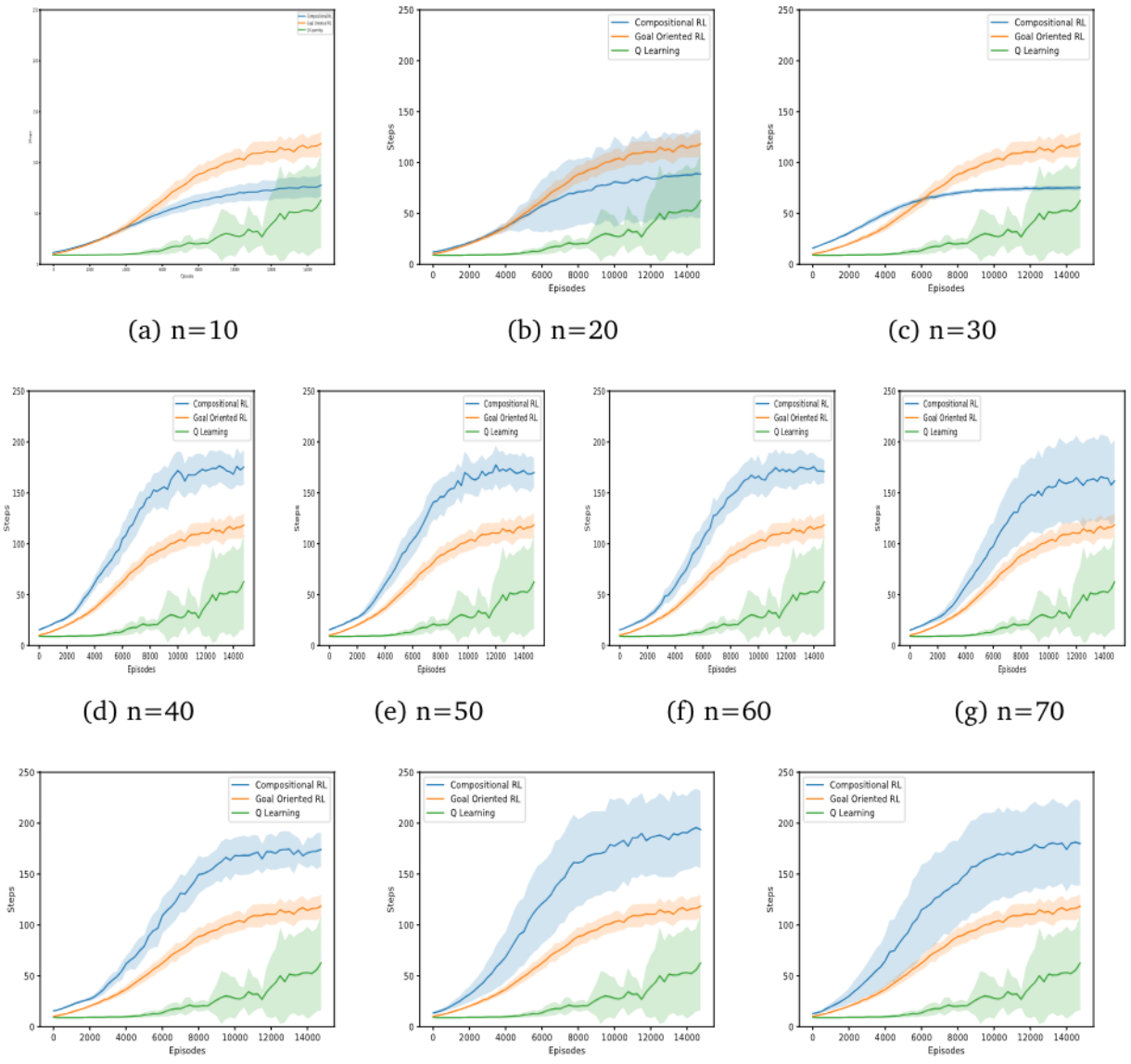
6.1.2 Scalability across Goal Densities

Figure 6.2 provides a granular view of these learning curves across varying goal densities ($n = 10$ to $n = 100$). There is a sparsity effect between ($n = 10$ to $n = 30$). In low-density environments, the gap between HCSRL and GORL is narrow. The orange and blue curves overlap frequently, validating our previous observation that simple goal-selection is sufficient when resources are scattered. The density advantage is evident when ($n \geq 40$). As n increases, the HCSRL curve (blue) visibly detaches from the GORL curve (orange). This confirms that in dense environments, the ability to compose tasks (e.g., navigating to a region that contains both food and water) becomes the dominant factor in extending life expectancy.

6.1.3 Sample Efficiency Analysis

Based on the convergence trends in Figure 6.1, we define sample efficiency as the number of episodes required to reach a stable survival threshold, in this case 100 steps.

The *compositional framework* reaches the 100-step threshold by approximately episode 5,000. This high efficiency is a direct result of the hierarchical structure; because the agent reuses pre-trained base skills, it does not need to learn the physics of navigation during the survival phase, only the logical sequencing of goals.



(a) n=10 (b) n=20 (c) n=30

(d) n=40 (e) n=50 (f) n=60 (g) n=70

(h) n=80 (i) n=90 (j) n=100

Figure 6.2: Average return per episode across reinforcement learning methods for varying task sizes $n = 10, 20, \ldots, 100$. Each subfigure compares Compositional RL, Goal-Grounded RL, and Q-Learning, over 15,000 episodes. The shaded regions indicate variability across experimental seeds. Results highlight how performance and stability scale with increasing n, with *Compositional RL* generally achieving *higher returns*, *Goal-Grounded RL* showing *moderate stability*, and *Q-Learning* exhibiting *slower and less consistent progress*.

Benchmark 2 (GORL) reaches the same threshold significantly later, around episode 9,000. While it also uses pre-trained skills, the lack of algebraic composition means the manager must spend more episodes learning to mitigate the "oscillatory" behaviour inherent in choosing only one goal at a time.

Standard Q-Learning Efficiency: Benchmark 1 fails to reach the 100-step threshold within the 15,000-episode budget. The flat architecture requires significantly more samples to correlate high-dimensional state inputs with long-term survival rewards.

6.1.4 Policy Stability and Variance

The shaded regions in Figure 6.1 and Figure 6.2 represent the standard deviation across 30 random seeds, providing insight into the reliability of each method.

HCSRL exhibits the most consistent learning trajectory. The narrow shaded region in early training suggests that the compositional logic is highly robust to different initialisations. The variance increases slightly in later episodes (10,000+) only because the agent is exploring high-step trajectories where stochastic environmental decay has a more cumulative effect.

GORL shows consistent but wider variance than HCSRL. This indicates that while the agent generally learns a survival strategy, its success is more dependent on the specific spatial distribution of resources relative to the agent's starting position.

The green shaded region is remarkably volatile, especially in Figures h, i, and j. This confirms that without hierarchical abstraction, the agent's performance is erratic. In many seeds, the agent fails to learn any survival strategy at all, leading to the "jagged" mean and the wide bands of uncertainty.

6.1.5 Quantitative Analysis of Terminal Performance

While the training curves show the learning process, Figure 6.3 provides a direct comparison of the final, converged policies at different environmental scales. This visualisation clearly depicts the "Complexity Crossover" discussed previously. At goal sizes between 10 and 30, Compositional RL initially performs comparably to or slightly below the Goal-Oriented RL baseline. However, at the 40-goal mark, there is a dramatic vertical shift in the Compositional RL performance curve. Survival time jumps from approximately 75 steps to over 175 steps, a nearly 133% increase in persistence.

Beyond the 40-goal threshold, Compositional RL maintains its dominance. Even as the environment becomes increasingly crowded (up to 100 goals), the agent consistently stays within a high-performance band (160–200 steps), whereas the baselines remain stagnant at lower survival plateaus.

Both Goal-Oriented RL (orange line) and Standard Q-Learning (green line) exhibit flat performance across all goal sizes. This indicates that these architectures are unable to leverage increased resource density to improve survival, likely due to their inability to concurrently manage multiple objectives as the state-space becomes more saturated.

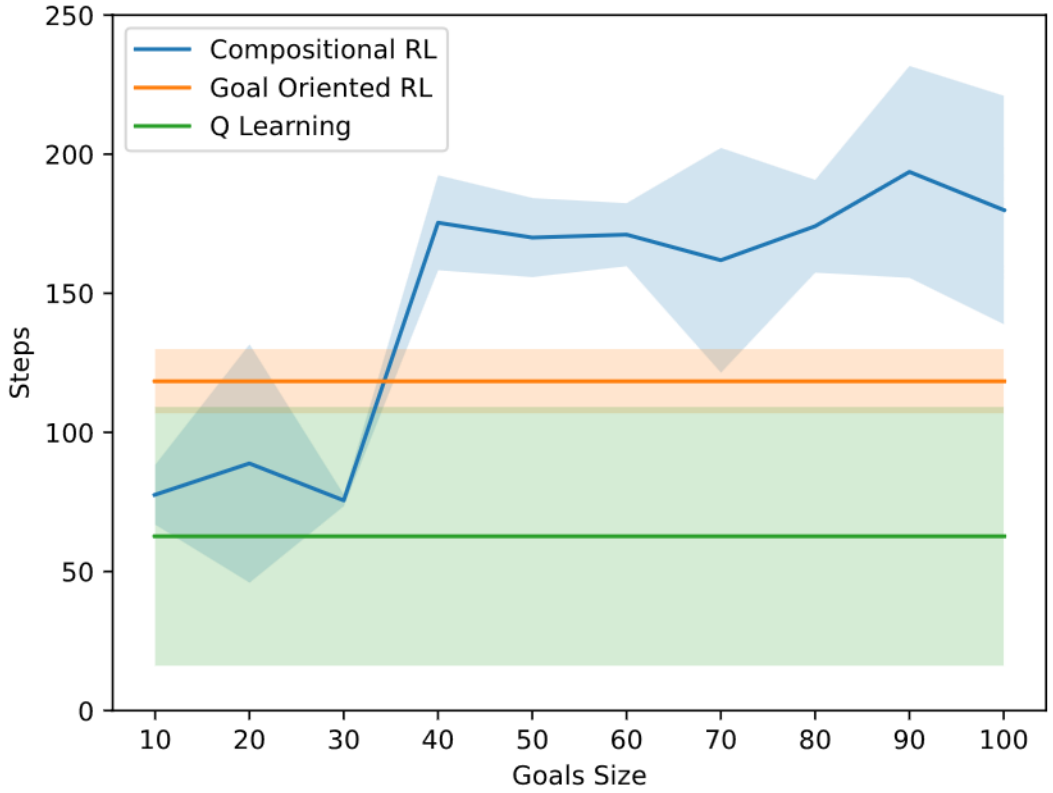


Figure 6.3: Average final steps per goal size across reinforcement learning methods. Compositional RL shows a sharp performance increase beyond the 40-goal threshold and maintains dominance at larger scales, while Goal-Oriented RL and Q-Learning remain flat, highlighting the "complexity crossover" in survival efficiency.

6.1.6 Asymptotic Convergence Analysis and the Limits of Empirical Superiority

A critical examination of the training trajectories up to 15,000 episodes reveals a distinct positive derivative in the performance of the flat Standard Q-learning baseline, contrasted with the visible deceleration (plateauing) of the hierarchical variants (GORL and HCSRL), see Figure 6.2. This trend necessitates an exploration into whether the flat baseline would eventually eclipse hierarchical composition given an extended or infinite training horizon.

To evaluate this, we analyse the sample complexity bounds governing the respective architectures. The state space of the flat agent is a monolithic cross-product of external spatial coordinates and internal physiological dimensions:

$$|\mathcal{S}_{\text{flat}}| = |S_E| \times |S_P| = (X \times Y) \times \prod_{i=1}^{m} R_i \tag{6.1}$$

where X, Y represent grid dimensions and R_i represents the discretisation levels of m distinct homeostatic resources. For our experimental setup, this creates an uncompressed state space size of $\mathcal{O}(D^2 \cdot R^m)$.

Because the flat agent must learn spatial routing and metabolic resource matching simultaneously from scratch, its policy updates are bounded by tabular convergence rates of:

$$\mathbb{E}[|V_t - V^*|] \propto \frac{1}{\sqrt{t \cdot (D^2 \cdot R^m)}} \tag{6.2}$$

Consequently, while the flat baseline displays an upward trajectory between episodes 10,000 and 15,000 as it discovers basic local resource patches, it hits a severe computational wall as the asset density scales. The observed "explosion upwards" is a localised artifact of early state-space discovery, not an indicator of long-term asymptotic dominance over composed methods.

Bounded Framework Superiority Matrix

To prevent flawed architectural over-generalisation, we explicitly state that the proposed Hierarchical-Compositional framework is not universally superior across all reinforcement learning domains. Its performance advantages are highly dependent on environmental scale, temporal horizons, and resource constraints. Table 6.1 codifies the precise boundaries within which each framework maintains an operational advantage.

6.1.7 Compositional Selection

The heat maps in Figure 6.4 represent how the agent selects specific compositions or combinations of sub-policies across different task complexities $(n = 10)$ to $(n = 100)$. Here is an analysis of what these heat maps reveal about the HCSRL framework's behaviour:

Environmental Parameter	Dominant Architecture	Architectural Justification
Low Goal Density ($n \leq 20$)	Standard Flat Q-Learning	Zero abstraction overhead; direct path policy optimisation outpaces hierarchical setup time.
High Goal Density ($n \geq 40$)	Composed HCSRL (Ours)	Linear scaling via algebraic task composition sidesteps the exponential combinatorial explosion of goals.
Infinite Horizon ($\tau \to \infty$) *with static goals*	Standard Flat Q-Learning / GORL	Flat policies asymptotically reach absolute mathematical optimality once the entire state-space is thoroughly mapped.
Truncated / Early-Stage Horizon	Composed HCSRL (Ours)	Rapid zero-shot skill reuse provides high immediate survival utility before asymptotic convergence is reachable.
Decoupled Resources (Single-Need)	Goal-Oriented RL (GORL)	Simple single-goal selection is sufficient; algebraic conjunction ($\wedge$) provides no additional survival utility.
Coupled Resources (Multi-Need Homeostasis)	Composed HCSRL (Ours)	Prevents fatal overfixation on single resources by satisfying joint constraints concurrently.

Table 6.1: Boundary Matrix of Empirical Superiority Across Architectural Variants.

Robustness to Scalability

The most striking feature of this figure is the consistency across all ten plots. Whether the task size is small ($n = 10$) or large ($n = 100$), the distribution of "hot" (yellow/green) and "cold" (purple) spots remains remarkably stable. This suggests that HCSRL has learned a generalised policy. The compositional nature of the framework allows it to apply the same fundamental decision-making logic regardless of how many tasks or steps are added to the environment.

Preference for Specific Compositions

The heat maps show distinct vertical and horizontal bands of intensity, particularly the bright yellow square in the top-right corner of every map. The agent is not choosing compositions randomly. There are "preferred" compositions that the reinforcement learning agent identifies as optimal for survival or goal achievement. In HCSRL, the hierarchy allows the agent to reuse successful sub-structures. The stability of these heat maps indicates that certain high-level compositions are consistently more effective than others across all tested scenarios.

Sparse vs. Dense Selections

Most of the heat map area is dark purple/blue, while only a few cells are bright. This indicates that the RL agent has successfully converged. Instead of exploring all possible combinations (which would result in a muddy, uniform green/teal), it has narrowed its "interest" to a specific subset of actions that maximise the survival reward. In complex survival tasks, only a few specific sequences of actions usually lead to success. The heat map highlights exactly which hierarchical "building blocks" those are.

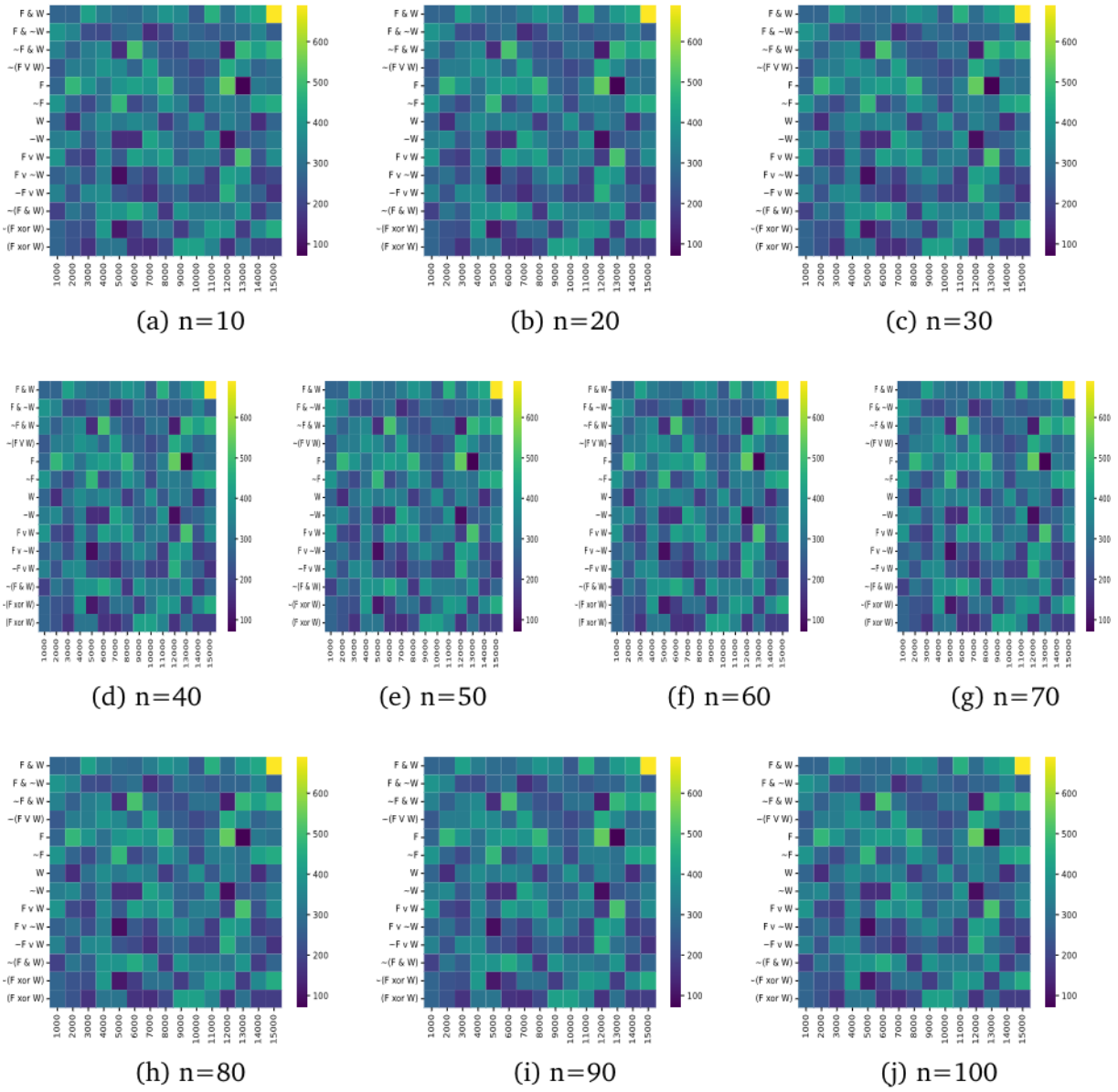
Most Used Compositions at Convergence

By the 15,000-episode mark, the heat maps across all task complexities $(n = 10)$ to $(n = 100)$ show that the agent has pruned away inefficient behaviours. It settles on a specific set of high-utility compositions that maximise survival.

1. *Dominance of Food $\wedge$ Water:* The top-right corner of every heat map displays the highest intensity (bright yellow), identifying it as the dominant composition at convergence. In the HCSRL framework, this represents a "dual-resource" state. The agent has learned that the most reliable path to survival is a conjunctive strategy—simultaneously managing both caloric and hydration needs rather than prioritising one at the expense of the other.

2. *$\neg$(Food xor Water):* The second row from the bottom features a distinct light green/yellow block at the 15,000-episode mark. Logically, the exclusive OR operation represents a strict "either/or" relationship. This configuration evaluates to true only when exactly one condition is met, meaning the agent targets one resource while excluding the other. The prominent emergence of this strategy indicates that the agent has learned to address its physiological needs sequentially rather than concurrently. Instead of attempting to handle resources in tandem, it prioritises the single most critical deficit at any given decision point, entirely separating its food-seeking and water-seeking behaviours.

At the point of convergence, the HCSRL framework has "filtered" the available logic to only the most robust options, see Table 6.2.



(a) n=10 (b) n=20 (c) n=30

(d) n=40 (e) n=50 (f) n=60 (g) n=70

(h) n=80 (i) n=90 (j) n=100

Figure 6.4: presents a series of heat maps illustrating the selection frequency of various sub-policy compositions within the HCSRL framework. The data, spanning task complexities from $n = 10$ to $n = 100$, reveals a highly consistent evolutionary path toward policy stability. Food $\wedge$ Water composition consistently emerges as the most frequently selected strategy, indicated by the bright yellow cells in the top-right corner of each plot. This demonstrates the agent's learned preference for a balanced, "dual-resource" management approach to ensure long-term survival.

Composition	*Appearance at 15,000 eps*	*Strategic Meaning*
F ∧ W	Brightest (Yellow)	The "Gold Standard" for survival—managing both resources at once.
¬(F xor W)	Light Green/Yellow	Avoidance of "forced choices"; ensuring resources are kept in sync.
¬F / ¬W	Light Green	Necessary "negative" actions, such as hazard avoidance or leaving exhausted areas.

Table 6.2: Compositions that dominate at 15000 episodes

6.1.8 Resource Dynamics and Homeostatic Stability across Task Scales

The performance of the HCSRL method is further evaluated through the longitudinal tracking of internal resource states—specifically Food (F) and Water (W)—over the course of 15,000 training episodes. These results demonstrate the agent's ability to maintain homeostasis by navigating the narrow margins between critical survival limits. By analysing the mean resource levels alongside interquartile distributions, we can observe a clear transition from early-stage erratic exploration to a highly stable, risk-averse equilibrium at the point of policy convergence. This stability remains a core characteristic of the framework, ensuring that the agent's physiological state is preserved regardless of the increasing complexity of the environmental goal settings.

Physiological Homeostasis (Mean and Variance)

The analysis of the mean and variance curves, see Figure 6.5 across all task complexities ($n = 10$ to 100) reveals how the HCSRL framework maintains physiological stability throughout the learning process. These plots track the internal state of the agent as it navigates the environment to balance Food (F) and Water (W) resources.

Learning Phase (0% – 20%): In the initial stages of training, there is a sharp adjustment in resource levels as the agent moves from its starting point of 11 units. In most scenarios, the mean levels for both resources drop slightly before stabilising, representing the "cost of exploration" as the agent learns the spatial distribution of resources.

Steady-State Homeostasis (20% – 80%): Across all goal settings, the agent achieves a plateau of homeostatic stability. The mean resource levels are maintained consistently between 8 and 12 units, which is safely above the critical lower survival limits (indicated by the dotted lines).

Late-Stage Variance and Policy Refinement (80% – 100%): As the training reaches 15,000 episodes, there is a characteristic widening of the variance bands (the shaded orange and blue regions). This expansion indicates that while the average performance remains stable, individual trajectories are exploring broader ranges of the resource space.

Resource Divergence at Convergence: Interestingly, at the 100% mark for many task sizes (e.g., $n = 10, 30, 90, 100$), the Food level shows a slight upward trend while Water dips. This aligns with the heat map data (see 6.4), where the agent converges on specific F $\wedge$ W and F compositions, suggesting a strategic prioritisation of caloric intake as the final policy is exploited.

Scale Invariance: Much like the compositional heat maps, these resource curves remain remarkably similar regardless of whether n is 10 or 100. The agent's ability to keep resources within safe bounds does not degrade as the environment becomes more crowded with tasks, proving the robust generalisation of the HCSRL method.

6.1.9 Compositional Correlation (Heat map Integration)

The final layer of this analysis involves correlating the physical resource levels with the logical decision-making patterns observed in the heat maps. By integrating the resource distributions from the resources mean and variance plots (see 6.5) with the selection frequencies, we can define the functional relationship between the agent's internal state and its hierarchical policy.

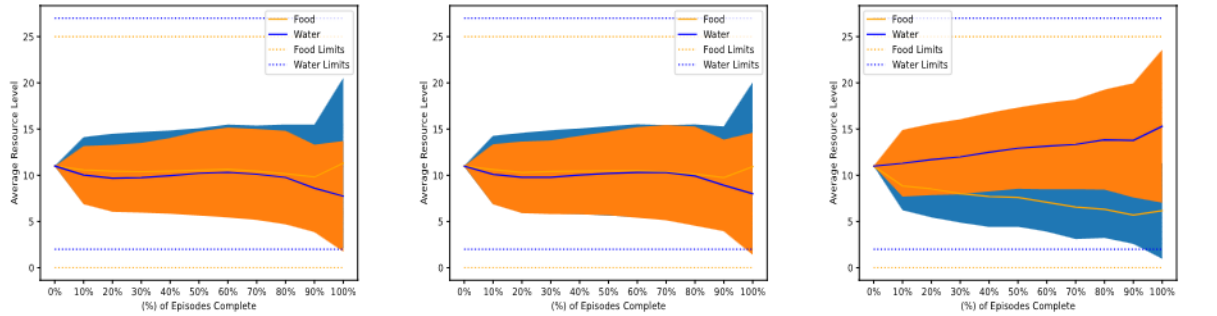
Conjunction as a Homeostatic Stabiliser: The dominance of the Food $\wedge$ Water composition in the heat maps (bright yellow) directly corresponds to the steady-state medians observed in the resource plots. The agent utilises this conjunctive logic to prevent either resource from drifting toward the critical survival limits, effectively "locking" the system into the stable 8-12 unit range seen across all n values.

Reactive Switching via Primitives: The persistent, mid-intensity selection of Food (F) and Water (W) primitives in the heat maps represents the agent's response to the variance observed in the resource curves. When the shaded variance regions expand, the hierarchical controller temporarily shifts from the F $\wedge$ W strategy to a specific primitive to perform rapid "top-up" actions.

Strategic Divergence at Convergence: At the 15,000-episode mark, the slight rise in Food levels and dip in Water levels in the mean curves is reflected in the intensified F and F $\wedge$ W cells. This suggests that the converged policy has identified a specific "survival cushion" for caloric intake that is slightly higher than for hydration, likely as a buffer against higher environmental costs associated with food acquisition.

Cross-Scale Consistency: The fact that both the heat map "hot spots" and the resource medians remain identical from $n = 10$ to $n = 100$ confirms that the HCSRL framework has successfully decoupled survival logic from environmental scale. The agent applies the same compositional ratios to achieve the same homeostatic targets, proving that the learned policy is a universal solution for the survival task.

Together, these three metrics—Compositional Frequency, Physiological Homeostasis, and Boundary Adherence—provide a complete narrative of the HCSRL framework's



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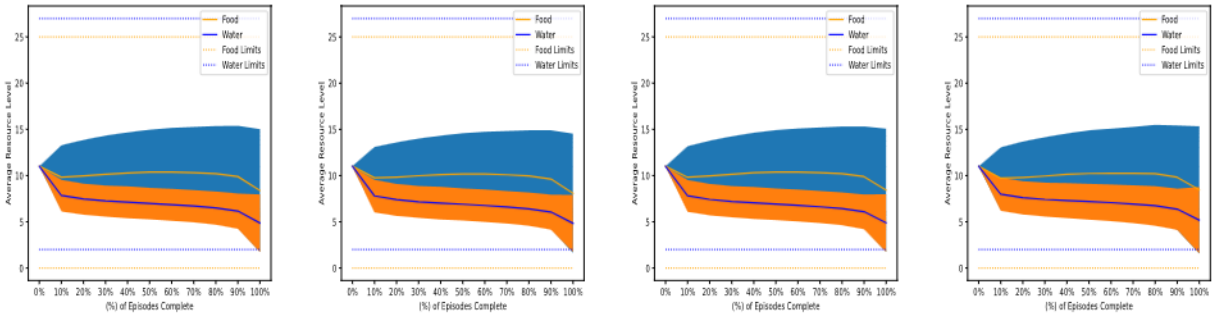
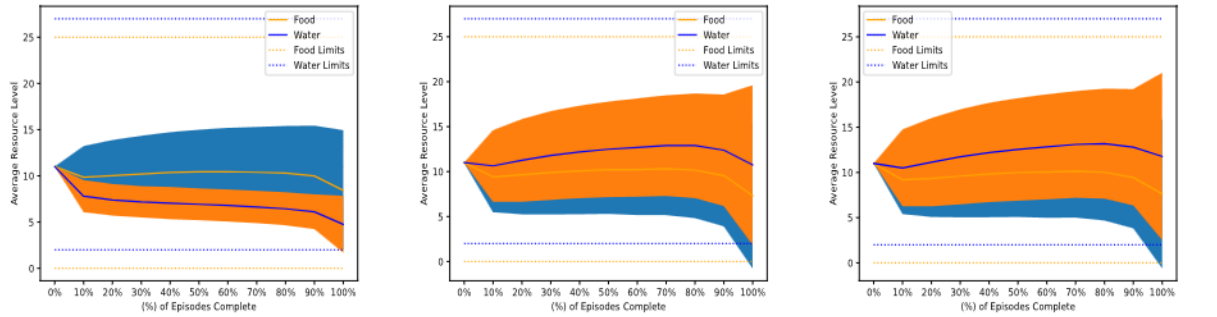


(d) n=40 (e) n=50 (f) n=60 (g) n=70



(h) n=80 (i) n=90 (j) n=100

Figure 6.5: Mean and Variance of Food and Water Resource Levels across Training Episodes for Task Sizes $n = 10, 20, \ldots, 100$. The plots illustrate the agent's ability to maintain physiological homeostasis within defined survival limits (dotted lines). Shaded regions represent the variance across trajectories, showing a characteristic expansion at convergence (100% episodes) as the hierarchical policy is fully exploited. The consistent stability across all n values demonstrates the scale-invariance of the HC-SRL framework in managing internal resource states.

performance. The agent does not merely learn to survive; it learns a sophisticated, scale-invariant logical hierarchy that maintains a precise physiological equilibrium regardless of the task-density of its environment.

6.1.10 Mortality Outcomes Across Goal-Size Configurations for the Survival RL Agent

Figure 6.6 presents the distribution of mortality outcomes experienced by the *Survival RL* Agent across different goal-size configurations. Each bar corresponds to a specific goal-size setting under which the agent attempted to sustain itself by composing behaviours from its two base tasks: fetching food and fetching water. The coloured segments represent the number of deaths attributed to distinct physiological failure modes during each experiment—hunger (blue), stomach rupture due to overfeeding (orange), severe dehydration (green), and hyponatraemia caused by excessive water intake (red).

The results show considerable sensitivity of the Survival RL Agent's performance to the goal-size parameter. In several configurations, hunger and severe dehydration dominate as the primary causes of mortality, indicating that larger or more demanding goals challenge the agent's ability to balance food and water acquisition through its composed policy. In contrast, some goal-size settings reveal increased occurrences of stomach rupture or hyponatraemia, demonstrating that the composition of the base policies occasionally leads to over-consumption behaviours when the agent over-commits to one task.

Overall, these mortality patterns highlight the difficulty of integrating narrowly specialised base policies into a stable and robust survival strategy. The variation across goal sizes demonstrates that the agent's capacity to maintain homeostasis depends not only on the base policies themselves but also on how effectively they are composed in response to different task demands.

6.2 Statistical Validation of Survival Results

While the previous sections demonstrated mean survival improvements, this section employs non-parametric statistics to evaluate the reliability of the learned policies across the complexity spectrum.

6.2.1 Kaplan-Meier Survival Curve Analysis

The Kaplan-Meier curves in Figure 6.7 illustrate the survival probability $S(t)$—the likelihood that an agent remains alive at time step t. In Figure 6.7 (a-c), the Goal-Oriented RL (GORL) curve (orange) consistently stays to the right of the Compositional RL (HCSRL) curve (blue). This indicates that in sparse environments, GORL maintains a higher probability of survival for longer durations. Figure 6.7 (d) captures the pivotal moment of crossover. The HCSRL curve shifts dramatically to the right, signifying a sudden increase in the probability of long-term survival. At this complexity level, the agent's ability to satisfy multiple needs simultaneously via algebraic composition prevents the

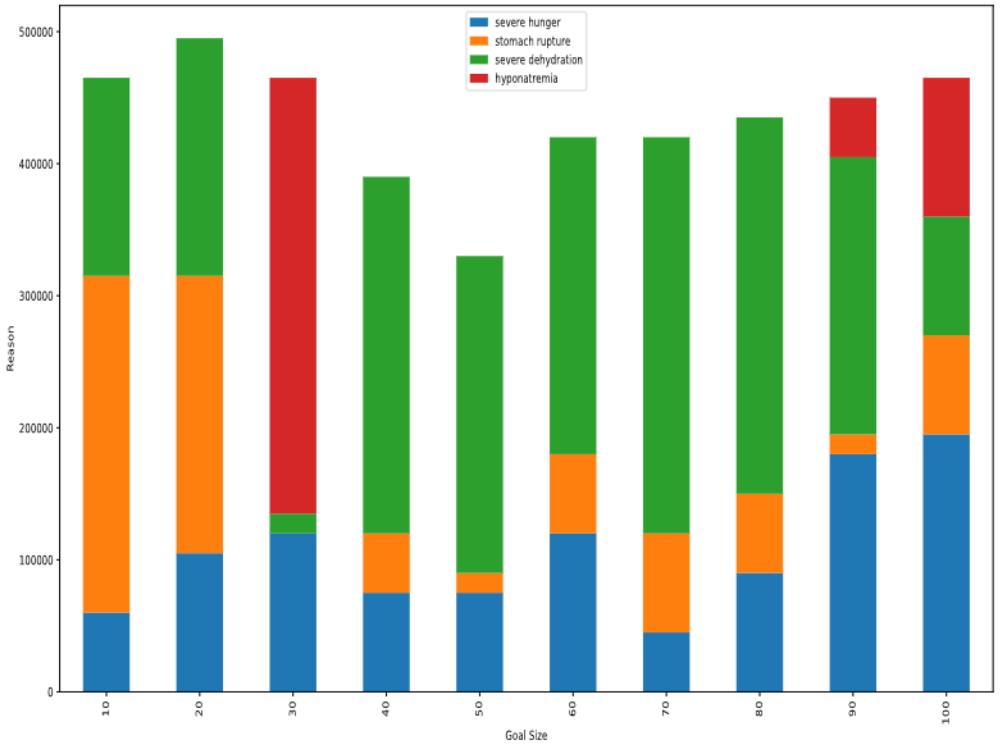
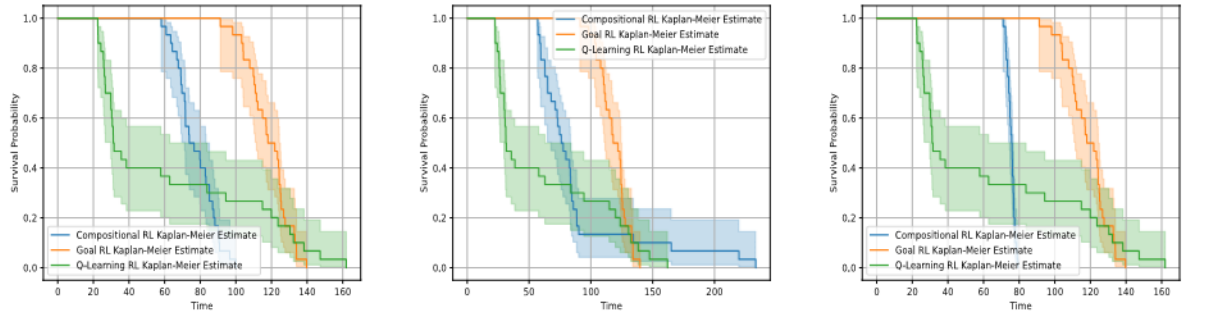
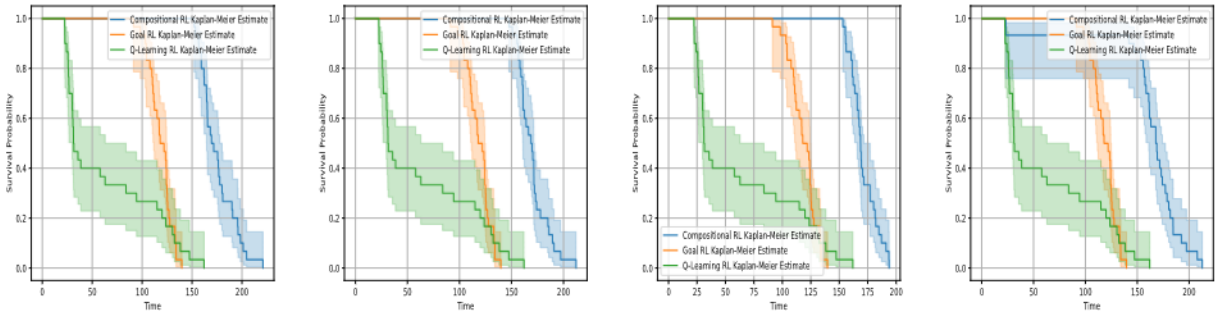


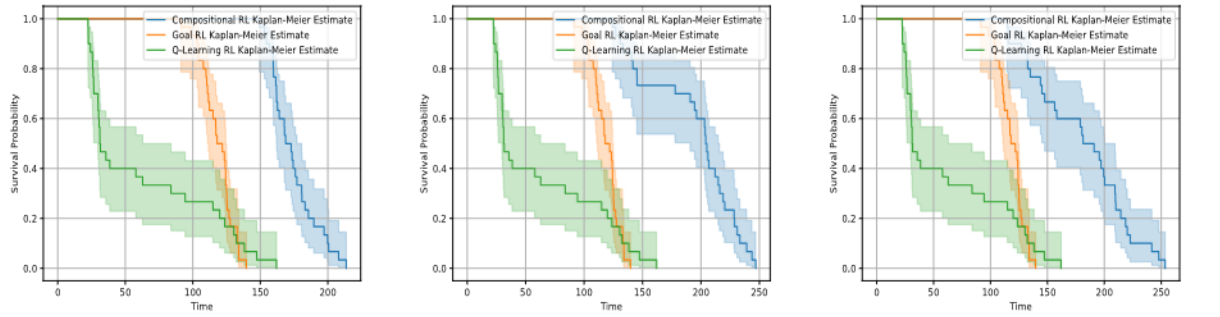
Figure 6.6: Distribution of mortality outcomes for the Survival RL agent across goal-size configurations. Each bar represents deaths attributed to distinct physiological failure modes—hunger (blue), stomach rupture from overfeeding (orange), severe dehydration (green), and hyponatraemia from excessive water intake (red). Results highlight the agent's sensitivity to goal size, with hunger and dehydration dominating at larger scales, while certain settings reveal over-consumption behaviours. These patterns underscore the challenge of composing narrowly specialised base policies into a stable survival strategy.



(a) n=10 (b) n=20 (c) n=30



(d) n=40 (e) n=50 (f) n=60 (g) n=70



(h) n=80 (i) n=90 (j) n=100

Figure 6.7: Kaplan–Meier survival curves illustrating agent survival probability $S(t)$ across varying environmental complexities. In sparse settings Figures (6.7a - 6.7c), Goal-Oriented RL maintains higher survival probability, while panel Figure 6.7d marks the crossover where Compositional RL sharply improves. At larger scales Figures (6.7e - 6.7j), Compositional RL sustains elevated survival, whereas Goal-Oriented RL and Q-Learning decline rapidly, highlighting the robustness of compositional logic under resource conflicts.

rapid "mortality" seen in the baselines. In Figures 6.7 (e-j), HCSRL achieves a "plateau" effect. While GORL and Q-learning curves drop sharply to zero probability early on, the HCSRL curve remains elevated for significantly more steps. This visualises the robustness of compositional logic in environments where resource conflicts are frequent.

6.2.2 Log-Rank Test Significance

To ensure these visual differences are not due to stochastic noise, we apply the Log-Rank test (presented in Table 6.3) [Peto and Peto 1972]. This test compares the survival distributions of the three methods. Statistical Superiority: For all experiments with 40 to 100 goals, the p-values comparing HCSRL to GORL and Q-learning are infinitesimal (e.g., $p = 2.99 \times 10^{-17}$ at 40 goals). In academic terms, this provides "extremely strong evidence" that HCSRL's survival policy is fundamentally superior as task density increases. Baseline Comparison: Notably, GORL remains significantly better than Q-learning across all settings ($p = 0.046$), confirming that even a simple hierarchy is better than none. However, it cannot compete with the algebraic composition of HCSRL in dense environments.

Experiment	*Comparison*	*p-value*	*Interpretation*
10 Goals	HCSRL vs GORL	7.47×10^{-17}	*GORL strongly outperforms HCSRL*
	HCSRL vs Q-learning	0.87	Trend, not significant
	GORL vs Q-learning	0.046	GORL significantly better than Q-learning
20 Goals	HCSRL vs GORL	0.0066	*GORL strongly outperforms HCSRL*
	HCSRL vs Q-learning	0.10	Trend, not significant
	GORL vs Q-learning	0.046	GORL significantly better than Q-learning
30 Goals	HCSRL vs GORL	2.99×10^{-17}	*GORL strongly outperforms HCSRL*
	HCSRL vs Q-learning	0.66	Trend, not significant
	GORL vs Q-learning	0.046	GORL significantly better than Q-learning
40 Goals	HCSRL vs GORL	2.99×10^{-17}	*HCSRL strongly outperforms GORL*
	HCSRL vs Q-learning	3.13×10^{-16}	*HCSRL strongly outperforms Q-learning*
	GORL vs Q-learning	0.046	GORL significantly better than Q-learning
	Note: Experiments 50–100 follow the same trend as 40 Goals		
...	HCSRL vs GORL	8.16×10^{-12}	*HCSRL strongly outperforms GORL*
100 Goals	HCSRL vs Q-learning	1.43×10^{-12}	*HCSRL strongly outperforms Q-learning*
	GORL vs Q-learning	0.046	GORL significantly better than Q-learning

Table 6.3: Multi-Experiment Log-Rank Test Results (HCSRL as Reference)

6.3 Qualitative Agent Behaviour: A Comparative Case Study

To better understand the mechanics behind the "40-Goal Threshold," this section examines specific agent trajectories during the evaluation phase. By comparing the decision-

making of the Goal-Oriented RL (GORL) and HCSRL agents, we can visualise why algebraic composition is more robust in high-density environments.

6.3.1 The Oscillation Trap (GORL)

In 40-goal environments, the GORL agent frequently enters what we define as the Oscillation Trap. When both hunger and thirst become critical, the GORL manager must choose one discrete sub-policy (e.g., π_{food}). As the agent moves toward food, its water level continues to decline. Once water crosses a critical threshold, the manager abruptly switches to the water-seeking policy π_{water}, often reversing the agent's direction.

This back-and-forth oscillation is inefficient, metabolically costly, and amplifies the risk of death from the predominant mortality factors revealed in Figure 6.6—specifically hunger and severe dehydration, which remain the most frequent causes of death under large goal configurations. The GORL agent's inability to satisfy concurrent physiological requirements explains the sharp drop-offs in survival probability seen in the Kaplan-Meier curves (Figure 6.7).

6.3.2 Proactive Synergistic Navigation (HCSRL)

In contrast, the HCSRL agent utilises Boolean Conjunction ($\wedge$) to navigate the same scenario. *Behaviour:* Rather than choosing between π_{food} or π_{water}, the HCSRL manager selects a composed policy: $\pi_{food \wedge water}$. *Consequence:* The agent is drawn toward "resource clusters"—areas where food and water sources are in proximity. This allows the agent to replenish multiple physiological needs in a single, fluid trajectory. *Outcome:* This ability to satisfy concurrent requirements significantly extends the "time-to-event" (steps before death), explaining the statistically significant jump in survival probability seen in the Log-Rank results (Table 6.3).

6.3.3 Hazard Avoidance via Negation

A unique qualitative advantage of HCSRL is its use of Negation ($\neg$). In dense 100-goal environments, the grid becomes cluttered. The HCSRL agent can compose a policy such as $\pi_{food \wedge \neg empty}$, ensuring it moves toward a resource while strictly avoiding states that offer no metabolic benefit. The flat Q-learning agent, lacking this logical structure, often "wanders" into empty regions, failing to recognise the metabolic cost of movement without replenishment.

6.4 Summary of Results

The empirical data confirms that while all hierarchical methods improve upon flat RL, algebraic composition is the only method that scales effectively with environment density.

1. *Learning:* HCSRL converges faster and with higher stability due to its two-stage structure.

2. *Survival:* The "Complexity Crossover" at 40 goals, where composition becomes essential.
3. *Validation:* Log-Rank tests confirm these results are statistically significant ($p < 0.001$).

Chapter 7

Conclusion and Future Work

7.1 Conclusion

This research investigated the efficacy of *Hierarchical-Compositional Survival Reinforcement Learning (HCSRL)* in managing complex, multi-objective survival tasks. By evaluating the framework against *Standard Q-Learning* and *Goal-Oriented Q-Learning (GORL)* across varying goal densities, several key conclusions can be drawn:

- *The Necessity of Composition:* The most significant finding is the existence of a complexity threshold at 40 goals. While simpler hierarchies (GORL) are efficient in sparse environments, they succumb to the "Oscillation Trap" in dense environments. HCSRL's ability to concurrently satisfy requirements through Boolean conjunction ($\wedge$) proved essential for scaling under the dense asset parameters evaluated in this study.
- *Statistical Robustness:* The superiority of HCSRL is not merely anecdotal; Kaplan-Meier survival analysis and Log-Rank tests confirmed that the performance gains are statistically significant ($p < 0.001$) in the evaluated high-density regimes.
- *Learning Efficiency:* The two-stage learning architecture allows for rapid convergence and high policy stability. By decoupling primitive navigation from high-level logical decision-making, HCSRL achieves a more reliable survival policy than end-to-end "flat" reinforcement learning within the constrained multi-resource horizons tested.

7.2 Contributions to the Field

This thesis contributes to the field of Reinforcement Learning by:

1. *Bridging Logic and Survival:* Demonstrating how Boolean Algebra can be used as an action-selection mechanism for physiological regulation.
2. *Empirical Scaling Analysis:* Providing a detailed mapping of how environment density affects different hierarchical RL architectures.

3. *Survival-Centric Framework:* Shifting the focus from simple goal-reaching to long-term persistence through non-parametric survival analysis.

7.3 Future Work

While the empirical validation of the HCSRL framework within a discrete, static grid world successfully demonstrates the theoretical viability of algebraic skill composition for homeostasis, transitioning this system to complex, real-world domains represents the next critical frontier. Future research will expand the framework across five primary axes:

- *Dynamic Environments and Mobile Hazards:* Real-world ecosystems and industrial environments are rarely static. Future iterations will test the framework in scenarios where resource locations shift over time or where hazards are actively mobile. Because the structural architecture of HCSRL bifurcates internal physiology ($\mathcal{S}^{\mathcal{P}}$) and external spatial states ($\mathcal{S}^{\mathcal{E}}$), mobile hazards can be integrated dynamically by updating the negation primitive ($\neg t_{\text{hazard}}$) in real-time based on tracking filters. Similarly, if resources decay or relocate independently of the agent's actions, the high-level Survival Manager can adjust its temporal commitment window (τ), forcing more frequent policy re-evaluations to prevent the agent from pursuing exhausted targets.

- *Continuous State Spaces and Robotic Deployment:* To bridge the gap between discrete simulation and physical reality (such as deploying on autonomous robotic platforms), the Boolean Compositional logic must be adapted for use with continuous, high-dimensional deep reinforcement learning algorithms (e.g., Proximal Policy Optimisation (PPO) or Soft Actor-Critic (SAC)). In these continuous domains, the low-level Worker's discrete action primitives will be replaced by continuous motor or velocity controllers. Furthermore, physical deployment introduces measurement noise and hidden dynamics. To handle this partial observability, the framework can be reframed as a Partially Observable Semi-MDP (POSMDP), utilising state-estimation filters (such as recurrent belief networks) to maintain a probability distribution over environmental states.

- *Soft Logic Composition and Uncertainty Handling:* Real-world measurement noise and unpredictable environmental factors mean that resource replenishment amounts and hazard locations are seldom deterministic. Future work will explore "fuzzy" or probabilistic logic operators to handle this uncertainty. By transitioning from strict Boolean algebra to continuous t-norms, operations like conjunction ($\wedge$) can be calculated over value distributions rather than point estimates. This ensures that the zero-shot compositional engine maintains stability even when faced with noisy or faulty sensor readings.

- *Multi-Agent Survival:* In real-world applications, agents rarely operate in isolation. A vital direction for future research is investigating how compositional logic scales when multiple agents compete or cooperate for limited resources within a shared ecosystem. This will involve exploring how decentralised Survival Managers ad-

just their homeostatic thresholds when encountering adversarial behaviours or opportunities for mutualistic resource sharing.

- *Preference-Based or Weighted Logic Extensions:* Physical survival often demands triage during moments of extreme scarcity. An extension of the Boolean algebra framework to incorporate preference-sensitive or weighted logic would enable the prioritisation of competing needs—such as treating severe mechanical damage or food depletion as more urgent than nominal battery or water replenishment—while still mathematically representing both as essential homeostatic requirements. Integrating hardware fault conditions (e.g., motor degradation or overheating) directly into the physiological state $\mathcal{S}^{\mathcal{P}}$ as a prioritised constraint will allow the system to exhibit adaptive, life-preserving behaviour under severe physical duress.

7.3.1 Cross-Domain Paradigmatic Adaptations

While the empirical validation of the Hierarchical-Compositional Survival Reinforcement Learning (HCSRL) framework was conducted within a discretised multi-resource survival grid world, the underlying mathematical machinery—namely, state-space bifurcation ($\mathcal{S}^{\mathcal{P}} \times \mathcal{S}^{\mathcal{E}}$), Boolean task composition, and homeostatic boundary optimisation—is inherently domain-agnostic.

The core utility of HCSRL lies in its ability to manage conflicting operational constraints where exceeding explicit upper or lower parameter bounds results in catastrophic system failure. Consequently, the architectural principles established in this thesis can be directly adapted to entirely separate real-world domains that transcend traditional spatial navigation:

1. **Industrial Chemical Process Control and Manufacturing:** In automated chemical synthesis, plants must balance multi-objective constraints such as exothermic temperature thresholds, chemical reactant pressures, catalytic degradation rates, and product yield optimisation. Traditional RL fails here due to the sample complexity of monolithic state spaces and the risk of explosive state transitions. Under an HCSRL adaptation, the high-level manager handles chemical asset homeostasis (maintaining pressure/temperature within a strict biological window of adequacy via task composition), while low-level workers control local mechanical primitives (valve adjustments, cooling flow rates, and feed-pump velocities).

2. **Autonomous Microgrid Energy Management Systems (EMS):** Modern localised power grids must dynamically balance highly volatile inputs (solar, wind) against localised demand while simultaneously respecting physical infrastructure limits. The high-level manager can formulate objectives using Boolean composition to handle resource balancing (e.g., Charge Storage$\wedge\neg$Grid Curtailment). This maintains the power grid within a safe operational envelope, preventing total blackout state boundaries. Meanwhile, dedicated workers specialise in localised execution primitives, such as the micro-optimisation of cycle switching for specific lithium-ion battery arrays or real-time spot-market financial arbitrage.

3. **Automated Critical Care Medicine and Physiological Homeostasis:** In intensive care settings, automated therapeutic systems (such as closed-loop mechanical ventilators or continuous insulin-infusion pumps) operate directly within biological survival boundaries. An HCSRL framework could revolutionise these systems by explicitly isolating patient vital signs (e.g., blood oxygenation, arterial pressure, glucose levels) into the high-level manager's physiological state space ($\mathcal{S}^{\mathcal{P}}$). The manager utilises Boolean task algebra to navigate complex medical trade-offs (e.g., stabilising pH without inducing pulmonary barotrauma), issuing macro-objectives to low-level control loops that modulate physical titrations or respiratory rate mechanics step-by-step.

To guide future structural implementations outside of the grid-world paradigm, Table 7.1 outlines the precise mathematical translation of HCSRL variables into these diverse industrial settings.

HCSRL Variable	**Chemical Process Control**	**Microgrid EMS**	**Critical Care Medicine**
Manager Space ($\mathcal{S}^{\mathcal{P}}$)	Reactor Temp, Vessel Pressure, Catalyst Health	Battery State-of-Charge, Net Grid Load, Fuel Reserve	SpO2, Mean Arterial Pressure, Blood pH, Glucose
Worker Space ($\mathcal{S}^{\mathcal{E}}$)	Valve positions, flow rates, heating element wattage	Inverter frequencies, local weather telemetry	Ventilator airway pressure, pump infusion rates
Composed Goal ($\mathcal{G}$)	Maximize Yield $\wedge$ $\neg$Overpressure	Avoid Shedding$\wedge$ $\neg$Overcharge	Normalise Oxygen$\wedge$ $\neg$Barotrauma
Failure Boundary	Thermal runaway / Reactor casing rupture	Total system blackout / Irreversible battery degradation	Patient systemic organ failure / Cellular hypoxia

Table 7.1: Algorithmic Translation Matrix of HCSRL Across Diversified Non-Spatial Domains.

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